FACSIMILIA

Game design: how the world works

This is the design plan for the game: every system, what it does, and the reasoning behind it. Each chapter opens with a short summary in plain English; the detail after it is for when you want the full picture.

(DECIDED) settled and either built or ready to build

(PROPOSED) drafted, waiting for your approval

(OPEN) not decided yet, or a value to tune later

Made from MECHANICS.md on 11 October 2026.

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In short

Your answers to the big design questions: sandbox with optional goals, realms that make their own choices but lean on history, full war and diplomacy, historical new realms plus revolts, culture and religion, player-chosen pacing for long games, tribes that start unorganized, and personal unions guided by history.

Answered by the designer in the Facsimilia Ledger. History is the guiding factor throughout: where these leave room, follow what really happened unless the player is involved.

- **Goal: sandbox with optional goals.** The game never has to end. It tracks a score and offers goals (hold a region, outlast rivals, rebuild Alexander's empire…) that the player may chase.
- **Other realms: their own choices, with history as a pull.** Realms decide for themselves but lean toward what they historically did, more strongly where the player isn't involved. (This matches the population engine's design: bots follow history, the player can bend it.)
- **War and diplomacy: full.** War must be declared before painting into a realm; peace deals end it; alliances and vassals exist. Conquest by painting (build step 7) works inside a declared war.
- **New realms: historical arrivals, revolts and civil wars.** New realms appear at their real dates and places (Galatia 278 BC, Parthia 247 BC, …). Revolts and civil wars follow history too, *unless* they happen in the player's lands or the player causes them; there the simulation decides.
- **Culture and religion: both**, with conversion and assimilation over time; together they drive integration (build step 6). Characters already carry naming cultures.
- **Pacing: the player chooses years per turn.** Games are meant to be long: a playthrough taking weeks or months is the target, not a problem. Design for depth over speed.
- **Frontier tribes start unorganized** (built): the Iberian, Gallic and Scythian peoples were confederations, not administered states, so their land starts with no provinces and follows the unorganized-land rules.
- **Personal unions: allowed, guided by history.** One ruler may inherit a second throne. The realms stay separate unless history merged them; if history renamed the realm on inheritance, the game does too. A union only splits again when the player is involved or history split it. (Not built yet: until it is, a monarch is still passed over for a second throne.)
- **Growth drivers: approved as drafted** (see Growth drivers).
- **Resources: every good simulated, shown by family.** All the goods are tracked; screens lead with about 12 families and open to the detail.
- **Resources follow history for other realms.** Mines, forests and soils under bots change the way they really did (Laurion's silver fading, Lebanon's cedars cut, Mesopotamia's soils salting); only the player can do better or worse than history.

- **Seasons: a yearly weather draw.** Each year draws its weather from each place's climate (good and bad harvests, droughts, floods); the sailing season limits sea trade.
- **Labour comes from the population, following history in full.** Workers are free, dependent or enslaved, as history had them. "History is king": people were traded, so the slave trade exists as it did, and slavery is outlawed when and where history outlawed it.
- **Land data: real datasets, with hand fixes** where they're wrong for 300 BC. Every source must allow the game to be sold: CC0, CC BY or public domain only, never "non-commercial" or "share-alike" (a test enforces it). The data is fetched by a script and the baked result is in the repository.

Next decisions (answered by the designer, 28 Sep 2026)

The designer's answers for sections 11-15 (a note overrides the chosen option where they differ). Playability (section 15) comes first.

1. **How battles are fought: A tactical battle map** (You move units on a field, like Total War. Much larger to build.)
2. **Who leads armies: Your family and courtiers** (Sons, brothers and trusted men lead; each has a skill; they can die, win glory, or rebel.) Designer's note: The royal family would die out sooner than later with this mechanic so make sure to add appointed generals, some are random events like "prince gaius imperius has met a roman of great prowess and has elevated him to general" except more roman englishy lol, and then some can be done by the player, it can be important people (a game mechanic that grants perks like in the civ games.
3. **How armies move: March by months within the year** (An army follows roads and rivers month by month; enemies can meet it on the way.) Designer's note: The route should be plottable also though, tying into the terrain update that will add features and obstacles
4. **Supply and winter: Supply lines and baggage** (Depots, convoys, foraging choices.)
5. **War at sea: Sea battles and blockades** (Fleets meet at sea, the loser's ships sink or are taken; a fleet off a port cuts its trade and supply.) Designer's note: Interactive battles just like land and air
6. **How you see a battle: A battle report** (The field, both armies by unit, the plan each chose, losses, the general's fate, a named battle ('the battle of Heraclea').)
7. **Fog of war: What your realm knows** (You see your neighbours, your trade partners and where your armies and envoys are; far lands are sketched, armies unseen.)
8. **Mercenaries: Named companies** (Real bands for hire from their homelands, costly, loyal only while paid; they may turn if unpaid (like Carthage's Mercenary War).)
9. **Veterans and army reforms: Both** (Units gain experience in battle and lose it to losses; techs and laws (like the Marian reforms) change how armies are raised and paid.)
10. **How other realms wage war: History's aims, sound tactics** (They pursue their history's aims but march sensibly, gather armies, defend their capitals and make peace when losing.)
11. **Rulers' abilities: Skills and traits** (War, diplomacy, stewardship and learning (0-20), plus traits (brave, cruel, pious, wise) that change them and how others see the ruler.) Designer's note: More than just 4 skills
12. **Do you play the realm or the family?: Choose at the start**
13. **Choosing heirs: Name an heir within the law** (You may name any son (or, where the law allows, daughter) as co-ruler; passed-over siblings may resent it.)
14. **Arranging marriages: You arrange them** (Offer a son or daughter to another court; a match brings opinion, alliances and claims.)

15. **A council of offices: A few offices** (General, treasurer, envoy, high priest, steward: each held by a character whose skill changes the matching part of the realm.) Designer's note: More than a few, about 8 is good.
16. **Republics and leagues: Elected magistrates** (You play the state; each year two consuls (suffetes, strategoi…) are elected from the leading families; their skill and ambitions shape the year; you lead one family among them.)
17. **Plots and ambitious nobles: Detailed intrigue** (Spies, schemes, secret alliances.)
18. **Faces for characters: Coin-style profiles** (Generated profiles in the style of ancient coins, by culture, age and sex; cheap and fitting.)
19. **Child rulers: A regent rules** (The mother or an uncle rules until the child is 16; regents may grab power.)
20. **A court screen: A family tree and court screen** (Ruler, spouse, heirs, courtiers, their ages, skills and traits; click to arrange marriages or appoint offices.)
21. **Events with choices: History's and chance events, with choices** ('Pyrrhus asks for ships', 'A comet is seen', 'The grain fleet is late': two or three choices with trade-offs.)
22. **History's great people: Born into their families when history had them** (If the family still exists, the person is born with fitting skills; otherwise not.)
23. **Great migrations: Moving peoples** (A people arrives from the map's edge at history's date as a wandering host that seeks land; it can be fought, bought off or settled.)
24. **New realms rising: History's plus successful revolts** (At history's date, if the land is weakly held, a new realm breaks away; any revolt that wins becomes a realm.)
25. **Religions spreading: They spread along roads and trade** (A faith appears where it began and spreads by trade routes, cities and missions; rulers can adopt, tolerate or persecute it.) Designer's note: History influences this.
26. **Peoples changing culture: Slowly, by rule, towns and schooling** (Provinces long ruled by another culture, with its colonies and cities, drift towards it over generations.)
27. **Great works: Yes, each once in the world** (Costly, slow to build, with lasting effects and score.)
28. **When the game ends: To today** (As the tech tree assumes: 2,300 years.)
29. **How strongly history pulls: A setting at the start** (Light, moderate or strong pull, chosen for each game.)
30. **What disasters do beyond deaths: More effects** (Plagues cut manpower and taxes and spread along trade; quakes destroy buildings; famines raise unrest; your choices (quarantine, relief) help.)
31. **Cities of their own: Cities with size and buildings** (Each town from Pleiades can grow through size tiers (village, town, city, great city); buildings go in cities; cities make crafts and trade, the countryside food.)
32. **Founding cities and colonies: Yes** (Pay silver and settlers to found a city or veteran colony; it grows, spreads your culture and loyalty, and anchors your rule.)

33. **Laws and policies: A set of laws** (Citizenship (who counts), land (big estates or smallholders), grain dole, slavery, levy, religion; each with trade-offs and some needing techs.)
34. **Changing your government: Through reforms or civil war** (Each form has its own rules; changing needs support or a civil war.) Designer's note: influenced by history
35. **Debasing the coin: Yes** (Strike more coins from the same silver: quick money, then prices rise, trade partners distrust your coin, and unrest grows.)
36. **Prices by place: Prices by market** (Each realm (or trade hub) has its own prices; goods flow from cheap to dear along the routes.)
37. **Borrowing: Lenders and other realms** (Temple treasuries, bankers and rich realms lend at interest; default angers them.)
38. **Building roads between provinces: Draw roads between cities** (They cost silver and years, speed armies, lower trade costs and extend administrative reach.)
39. **Mines running out: Yes, slowly** (Rich veins thin over centuries; new techs and prospecting find more.) Designer's note: Primarily according to history
40. **Slave revolts: Yes** (Where many are enslaved and treated harshly, revolts break out and can grow into armies.) Designer's note: Outside of players realm, only if history permits
41. **Learning to play: Both**
42. **An outliner of what needs attention: An outliner on the side** (Armies, sieges, wars, offers waiting, provinces near revolt, research done; click to go there.) Designer's note: collapsible
43. **Time passing: Month-by-month time with pause** (Needs monthly marching and events.)
44. **More map views: Culture, religion, loyalty, wealth, people, diplomacy** (Diplomacy shows allies, vassals, rivals and wars from your view.)
45. **Hover information: A small card** (Province, owner, people, culture, loyalty, garrison, the nearest town.)
46. **Comparing realms: A realms table** (People, land, silver, income, army, fleet, techs, score, sortable; plus graphs over time.)
47. **When your game ends: A history of your realm** (Score by category, a map of your peak, your rulers and great deeds, graphs of people and wealth over the centuries.) Designer's note: Alongside this should be a score of how close or far from history the player ended up
48. **Music and sound: see note** Designer's note: Create a second music ledger with the ideas and what kind of music you think we should have alongside what it will be used for, and I will make it myself.
49. **The look changes with the ages: Yes, by era** (Antique, late antique, medieval, early modern and modern themes: fonts, colours, ornaments.)
50. **How fast a turn must be: Under 3 seconds** (Room for more simulation.)
51. **Changing the game's data: Documented data files** (A guide to each file so you (or others) can add realms, events or techs, checked on load.)
52. **Envoys and relations: Envoys and gifts** (Send envoys (characters) to improve relations, sound out alliances, demand tribute or insult; gifts of silver or goods.)

53. **Client kings and leagues: Client kingdoms, leagues, protectorates** (Clients keep their kings but follow your foreign policy; leagues share a war policy; protectorates pay for protection.)
54. **Civil wars: Yes** (A rival claimant, a rebel general or a split senate takes part of the realm and armies; the war ends when one side wins.)
55. **What 'playable' means first: Playability: outliner, tooltips, map views, tutorial (section 15)**

Playable decisions (answered by the designer, 26 Sep 2026)

The 32 "Playable" questions in the Ledger. Where the designer's note adds to the chosen option, the note wins.

1. **Borders' licence: redraw the 300 BC borders ourselves** from open sources, so no GPL data ships with the game.
2. **Turns: 1, 5, 10 or 25 years**, default 1 (built).
3. **Long turns stop at major events** concerning the player (built).
4. **Money: each realm's own coin** (denarii, drachmae, shekels…), with silver weight behind them for comparison.
5. **Taxes: one rate you set**, with optional per-province rates.
6. **Running out of money: historical remedies**, offered according to the realm's situation at the time (borrow from temples, debase the coin, confiscate, sell offices or land…).
7. **Goods: every good matters from the start** (built).
8. **Trade: trade routes from the start**, not a pooled market.
9. **Armies: named armies based in provinces**, each with its own list of units (so many elephants, so many pikemen, so many scout ships).
10. **Units: every culture's own units.**
11. **Manpower: historical shares** of people who could serve (built).
12. **Conquest: sieges over years**, decided by the defending troops in the province and the realm's overall Might against the attacker's troops in its provinces and overall Might.
13. **Limit per turn: reach and supply** (built: reach).
14. **Sea: ships needed to cross** (built).
15. **War: declare on anyone, but pretexts matter.** Inspired by Stellaris' casus belli system, without copying it.
16. **Peace: terms by war score**, plus war exhaustion so peace can be forced. Defenders (realms and even single provinces) may offer things to appease an attacker, but only when the attacker is the player or history allows it.
17. **Alliances and vassals: in the first playable version.**
18. **History's pull on other realms: both** events at their dates and goals they pursue (built: goals).
19. **Other realms treat the player as history would** (react to threats, rivals, opportunity).
20. **Cultures: every people**, finely divided.
21. **Integration: historical pace** (built).
22. **Revolts: unrest builds, then revolt** (built).
23. **City names: the Pleiades gazetteer** (CC BY).

24. **Technology: a tech tree you choose from**: an advanced, complex web of techs that unlock as time passes but must still be researched.
25. **Empire size: administrative reach**, plus corruption and rivals, in a historical but playable way.
26. **Score: people, land, wealth, culture and dynasty**, shown separated by category (like Stellaris' final score).
27. **Goals: a starter set** (built).
28. **Plagues, famines and disasters: history's own plus chance.**
29. **Difficulty: none.**
30. **Playable realms: any realm on the map.**
31. **Province panel: essentials plus detail tabs.**
32. **Buildings: a core set** to begin with.

Core interaction principle: the brush stays authoritative

IN SHORT

You shape the world with one tool: a round brush. Conquest, provinces and anything added later all use that same brush, instead of menus or separate editors.

The existing left-drag-to-paint / right-click-to-clear ownership brush (`src/World/MapView.cs`, `PaintRadius`, the proposal overlay) is the one mechanic everything else builds on top of, not around. Every new layer described below reuses that same brush code path (radius stamp, drag-paint, dirty-cell tracking) rather than introducing a different interaction paradigm (no click-to-select-a-region UI, no modal province editor). New layers just give the brush a different target to paint onto.

Layers

IN SHORT

The map is stored as stacked layers over the same 45-million-cell grid: who owns each spot, which province it belongs to, and how many people and resources are there. A province's numbers are always added up from its cells, so redrawing a border can never break anything. “Provinces as built” describes what is in the game today.

Three per-cell layers, all the same 8192x5476 shape as the existing ownership grid:

1. **Ownership** (`OwnershipGrid.cells`, existing) — which realm holds this cell. Painted directly today (immediately, no commit step yet).
2. **`province_id`** (new) — which province this cell administratively belongs to. `0` = unorganized. Brush-editable in a "province mode" that targets this layer instead of ownership, with a selected target (an existing province of yours, to expand it, or "new province", to found one).
3. **Density fields** (new) — population and raw-resource presence. Population is **live, not baked**: it's `C * H_final^k`, recomputed each population tick from the blended simulated and historical heatmaps (see Population & settlement engine, below). HYDE (see Might, below) provides the historical side of that blend and seeds the 300 BC starting state. Raw-resource presence is still baked at world-gen from terrain data already loaded for the terrain shader. Not normally player-painted (geography/history-driven), though the same brush code could hand-adjust them in a dev/authoring context.

Provinces never store population/resources as an opaque blob — a province's stats are always the sum of the density fields over whatever cells currently carry its `province_id`. This is what makes provinces safely brush-editable: resizing a province's boundary just changes what it sums over, nothing needs manual splitting/merging logic. (DECIDED)

Invariant: ownership on organized land lives on the province, not the cell

For a province (`province_id != 0`), the *province* is the authoritative owner (`realm_of_province[province_id]`) — every member cell is owned by whoever owns the province, always wholly, never partially. `OwnershipGrid. cells` still holds a per-cell realm id (unchanged, so the renderer and existing tests keep reading it exactly as cheaply as today), but for organized cells it's a write-through cache: whenever a province's owning realm changes, every one of its member cells gets bulk-written to match in one pass — the same bulk-write pattern `_seed_real_civs` already uses for the political mask — never decided cell-by-cell.

For unorganized land (`province_id == 0`), there's no province to defer to: `OwnershipGrid.cells` is the sole, direct source of truth, exactly as it works today. Ownership there still changes cell-by-cell via the brush, independent of any province. (DECIDED)

Provinces as built (decided, built — build step 1)

Code: `src/World/Provinces.cs` (the layer and records), `src/World/ProvinceGenerator.cs` (the 300 BC provinces), `src/World/MapView.Provinces.cs` / `MapView.ProvinceBrush.cs` (drawing, selection, brush), `src/UI/Hud.Provinces.cs` (panel and map modes). Test: `ProvinceTest`.

- **Layer.** One province id per cell (`ushort`, 0 = unorganized), saved as `provinces.bin` beside the grid; names and owners go in `state.json`. Stats are always summed over cells (area now; population from the engine).
- **The 300 BC provinces.** `data/province_seeds.json` lists 245 historical regions c. 300 BC (Latium, Campania, Attica, Babylonia, Media, the Egyptian Delta, Meroe, Celtiberia, Arvernia, Taurica…). Each one on a realm's land becomes a province of that realm; provinces grow from their seeds over the realm's own land until they meet, then their shared borders get a gentle meander. Wherever land is more than ~150 km from any region, an extra province is added and named from the nearest region ("Northern Media"). Unseeded islands under ~1,500 km² join the nearest province. Result: two-thirds of them historical names, median ~21,500 km², about 190 provinces in all. The frontier tribes (Iberia, Gaul, Scythia) start unorganized, with no provinces (see Design decisions).
- **On the map.** Province borders are thin and fainter than realm borders, getting stronger as you zoom in; province names appear from about 2x zoom. Unorganized owned land shows paler. The selected province is lightened and outlined in gold.
- **Map modes** (HUD buttons or keys 1-3): *Inspect* (click selects a province, drag pans), *Provinces* (the province brush), *Conquest* (the old proposal brush, not yet playable).
- **Province brush.** Same round brush as conquest, a constant size on screen. Left-drag adds the player's own land under it to the selected province (one of the player's); right-drag takes land out of any province, leaving it unorganized. It never touches other realms' land. *New Province* founds an empty one to paint into; a province whose last cell is painted away disappears. The player can rename their own provinces in the panel.
- **Invariant.** `ProvinceMap.SetRealm` hands a whole province to another realm and rewrites every member cell's ownership in one pass.

Unclaimed vs. unorganized land — two different things

In short

Unclaimed land has no owner at all. Unorganized land has an owner but no province yet, so it is weaker: no taxes, only dirt roads, just a local militia. That weakness is the reason to organize land you take.

Easy to conflate, so pinning it down: **unclaimed** land (`owner_id == 0` in the existing ownership grid — today's "wild" cells) has no owning realm at all. **Unorganized** land has a realm owner, just no province yet (`province_id == 0`). They share one mechanic and diverge on everything that depends on having an owner to credit output to.

Population is a base geographic fact, independent of ownership — the density field (see Layers, above) exists on *every* land cell, unclaimed or not. Unclaimed land has its own (typically sparse) native population, same as unorganized or organized land does. Local militia strength derives from that population everywhere it applies: unclaimed land resists incursion the same way unorganized land resists conquest, scaled by however populated it actually is. What differs is output — unclaimed land has no owning realm to credit raw resources to, so **it produces nothing for no one**, even though the same coal/ore/water/flora deposits are physically present in its density field. The moment it's claimed and becomes unorganized territory of some realm, that realm starts drawing the base-rate raw output described below. This also refines the earlier assumption that settling unclaimed land is an unconditional freebie: it's cheap in practice (native population, and therefore militia, is usually sparse) but not literally free — it still resolves against whatever local defense exists. (DECIDED)

Newly gained land (settled or conquered) defaults to `province_id = 0`. It's real territory, but capped well below an organized province:

Capability	Unclaimed (no owner)	Unorganized (owned, no province)	Organized (in a province)
Population	Yes — sparse, native	Yes	Yes
Raw resource extraction (coal, ores, water, flora/fauna)	None — no owner to credit it to	Yes, base rate, credited to owner	Yes, improvable with infrastructure
Resource processing	No	No	Yes
Roads	None built; at most a natural/trade trail passing through	Dirt & gravel only	Full tiers as tech allows
Electricity routes	Cannot pass through	Cannot pass through	Allowed once unlocked

Capability	Unclaimed (no owner)	Unorganized (owned, no province)	Organized (in a province)
Military / defense	Population-scaled local militia, resists incursion	Population-scaled local militia, non-deployable	Full army stationing + logistics
Taxation	None (no owner)	None	Yes
Named settlements	Suppressed, same highly-populous exception applies	Suppressed, same highly-populous exception applies	Full, plus founding new ones

Deliberately weak defense on both unclaimed and unorganized land is the incentive to actually organize conquered/settled territory rather than leave it as paperwork. (DECIDED)

Settlement naming threshold — percentile rank, with hysteresis

Supersedes the earlier "10% of the world's largest settlement" rule (and the flat 50,000 placeholder before that). Both of those measured *size*; what the rule actually needs to protect is *map readability*, and that's a question of how many labels are on screen, which a percentile controls directly. Full rules, including where the populations being ranked come from, are in **Population & settlement engine**, below. Short version: a population cluster gets named when it enters the top **2-5%** of active clusters, and keeps its name until it falls out of the top **20%** or hits zero. This is the "highly populous exception" the capability matrix above refers to. It applies the same way on unclaimed, unorganized, and organized land.

(DECIDED)

Population & settlement engine

Three stages, run in this order every population tick:

1. **Blend** the simulated and historical heatmaps into one `H_final`.
2. **Translate** `H_final` into a target population per node, and move each node's actual population a limited step toward it.
3. **Rank** clusters by population, and name or un-name settlements.

1. Historical gravity: the attractor field

Two heatmaps are kept, both normalized to 0.0-1.0 per node:

- `H_sim` (simulated heat) is driven by gameplay: food, water, trade routes, infrastructure the player or AI builds, war damage.
- `H_hist` (historical target heat) is a static, invisible reference, never shown to the player. It carries e.g. a heavy spike over Rome or the Nile Delta in antiquity.

```
H_final = (1 - α) * H_sim + α * H_hist        α ∈ [0.15, 0.20]
```

A convex blend of two 0-1 fields stays in 0-1, so no renormalization is needed. Since `H_final ≥ (1 - α) * H_sim`, a player who builds heavily in a historically barren region always keeps at least 80-85% of their own heat there. `H_sim` overpowers the historical pull, which is what makes alternate history possible. (DECIDED)

Source of `H_hist`: HYDE, not a hand-authored map. (decided) HYDE is already the decided source of the population density field (see Might, below), and it ships gridded population snapshots from 10,000 BCE to the present (the game uses HYDE 3.2.1, which ends at 2017). That's exactly a sequence of historical target maps, baked onto this projection by `build_population_mask.py`. So `H_hist(year)` = the HYDE snapshot for that year, normalized to 0-1. Between HYDE's snapshots (per-century from 0 AD to 1700, per-decade to 2000, yearly after), interpolate linearly so the pull moves smoothly instead of jumping at each keyframe. HYDE has no 300 BC snapshot: before 0 AD it steps by 1000 years. The 300 BC keyframe is derived from 1000 BC and 0 AD per node, assuming steady exponential growth in between (39.4 million people on the map, against 26.1 million at 1000 BC and 48.2 million at 0 AD). At the 300 BC start, `H_sim` is seeded from `H_hist` itself, so turn one is historical by construction and divergence only comes from play.

The slow pull comes from population feedback into `H_sim`. (decided) On its own the blend is instantaneous and memoryless. An ignored but historically important region would get `α * H_hist` worth of heat once and then stay there forever, because nothing feeds `H_final` back into `H_sim`. So `H_sim` is partly a function of the previous tick's population, since people attract trade, labor, and markets:

```
H_sim(t+1) = normalize( drivers(t+1) + β * (Pop(t) / max Pop(t)) ^ (1/k) )
```

Here `drivers` is the gameplay term (food, water, trade, infrastructure, war) and `β` is how strongly existing population draws more. Taking the `1/k` root converts population back to heat scale, so the feedback carries population's momentum without adding concentration of its own. `k` stays the only concentration knob. At the 300 BC start the feedback term equals `H_hist`, so the historical starting state is a fixed point: nothing drifts until play changes the drivers. The small historical boost from `α` then compounds into a real drift toward history, and heavy, sustained player investment in `drivers` can still beat it. That's the "realist alternate history" behavior. Explicit relaxation of `H_sim` toward `H_hist` was the alternative. It was rejected because it drags on the simulation directly instead of acting through population.

`β = 5.0`, together with population inertia `μ = 0.065` (stage 2). Both are calibrated to real founded cities on the real HYDE grid. (decided; `μ` refitted from 0.0625 once regional totals were built, see Fit) Feedback alone can't make history slow to bend. The update is a contraction whose per-tick factor is at most `1 - α`, so any change in drivers settles within a few years whatever `β` is. A simulation showed a barren node turning into a major city within 0-1 years. The timescales come from population being a stock (see stage 2). The pair `(β, μ)` is then fitted to how long real non-capital cities took to climb the rankings. Capitals get their own calibrated bonus on top (see Capital growth bonuses, below).

Historical target. Years from a city's founding until it entered the world's top 50 and top 10 cities by population. The figures are rough estimates from Chandler (*Four Thousand Years of Urban Growth*) and Modelski. The sample is cities that did make the list; most foundings never do. "Capital" means the city was a country's capital from founding or within its first few decades.

City	Founded	Capital?	→ top 50	→ top 10
Alexandria	331 BC	yes	~20	~50
Seleucia-on-Tigris	305 BC	yes	~25	~55
Antioch	300 BC	yes	~50	~100
Pataliputra	490 BC	yes	—	~190
Constantinople (refounded)	330 AD	yes	~10	~60
Baghdad	762	yes	~10	~30
Samarra	836	yes	~8	~15
Edo	1603	yes	~20	~50
St. Petersburg	1703	yes	~50	~160
Calcutta	1690	no	~110	—
New York	1624	no	~190	~226
Philadelphia	1682	no	~150	~210

City	Founded	Capital?	→ top 50	→ top 10
Chicago	1833	no	~37	~57
Melbourne	1835	no	~50	—
Los Angeles	1781	no	~140	~170
Shenzhen	1980	no	~20	—
Average, capitals			**24.1 → 24**	**78.9 → 78**
Average, non-capitals			**99.6 → 100**	**165.8 → 166**
Average, all			*59.3 → 60*	*105.6 → 106*

Targets are the averages rounded to the nearest even number. The all-cities average (60/106) was the first calibration. It blended two different populations: capitals founded by decree and organically grown commercial cities. So the base engine is now fitted to **non-capitals (100/166)**, and capitals reach **24/78** through their own bonus.

Fit. `python3 tools/calibrate_population.py --hyde` simulates every land node of the real baked 300 BC keyframe (284,626 nodes), `k = 2`, yearly ticks, history held at 300 BC. It measures the median populated node given drivers matching the world's best site and held there. Ranks are within the map's extent, not the whole planet, so the map region's rank-size curve is what matters. A coarse-then-fine grid search gives `β = 5.0`, `μ = 0.0625` → **top 50 at year 102, top 10 at year 168**, the closest fit found on one map-wide pool.

Refit on the regional engine (built, 25 Sep 2026). Once each geographic region kept its own total (see Growth drivers), a new city drew only on its own region, and the same constants gave 102/172 years. A grid search on the regional engine itself, on the real grid with the 38 regions, kept `β = 5.0` and moved `μ` to **0.065**: **top 50 at year 98, top 10 at year 165**, against the historical 100/166, closer than before. `src/Tests/PopulationHydeTest.cs` holds this fit and checks the engine reproduces it. `tools/calibrate_population.py` still models the single pool and the toy world.

HYDE's top is flat: at 300 BC the #10 node holds ~20,000 and the #50 ~15,000, so a city that passes #50 is close to #10. Getting the historical 66-year gap between the two needs a city whose growth slows sharply as it nears the top. A large `β` does that: existing population, not drivers, dominates heat, so a newcomer's pull weakens against the established giants. The fast `μ` then gets it to the top 50 on time.

The constants were first fitted on a synthetic toy world (5,000 nodes on an ancient rank-size curve, population ∝ rank^-0.6): `β = 0.25`, `μ = 0.0115`, 96/166 there, but 143/165 on the real grid. The toy world is kept only as a fast test fixture.

Behavior at these values, measured on the real grid (the old toy-fit values in brackets):

Scenario	Result
Barren median node, drivers maxed and held indefinitely (player only; see Historical ceiling)	Top 50 at **102 years**, top 10 at **168 years** (98 and 165 on today's regional engine). Levels off at ~21% of the largest city's population, the map's #8 [62%, #2]
Same city, investment abandoned after it matures	Half its growth (measured on a log scale) is gone in ~73 years [104]
6th-largest historical city loses **all** its drivers for 30 years	Loses ~37% of its population [49%] and is back to 90% of its former size ~72 years after the war ends [191]. Established cities are sturdier: their own population holds their heat up

History's pull at these values comes from `α`, from the large `β` (established places keep their weight), and, for everyone except the player, from the hard Historical ceiling below.

2. Heatmap-to-population translation

```
Target_node   = C * H_final ^ k
Pop_node(t+1) = Pop_node(t) * (Target_node / Pop_node(t)) ^ μ        μ = 0.065 / year
```

Population is a **stock**, not recomputed from scratch each tick. People can only be born, die, and migrate so fast. Each year a node closes ~6.5% of the gap to its target, **measured on a log scale**: growth and decline are percentage rates, like real demography. A town 100x below its target grows ~33% a year, a gold-rush boom that fades fast as it closes in. A city 10% below its target grows ~0.7% a year. The log-scale form is required, not stylistic. A linear step (`Pop += μ(Target - Pop)`) grows fastest at the very start, which makes a founded city's climb from top 50 to top 10 take at least ~2.6x as long as reaching the top 50, versus ~1.8x historically. Compounding growth matches the real curve. It's what makes bending history a long-term strategy rather than something one build order does. Each node's own `Pop` is the value everything else reads (Might, taxes, naming, the feedback term above). (DECIDED)

Populations are whole people. Starting and seed populations come from HYDE; see Starting and seed populations, below. (DECIDED)

This is a power law, not exponential decay, but it has the intended effect: a higher `k` crushes middling heat far harder than peak heat, so population funnels into the hottest nodes.

- `C` **(carrying capacity, era multiplier)** scales with technology unlocks: about **150,000** for ancient megacities at the 300 BC start, and **tens of millions** by the industrial and space eras. It's implemented in world-total form (see below), so those per-city numbers are calibration checks on the hottest node, not a literal constant.
- `k` **(urbanization exponent)** rises with era. `k = 2` in ancient and agrarian eras spreads population out. `k = 4-5` in modern and interstellar eras simulates metropolitan concentration.

(DECIDED)

Two consequences worth knowing before tuning:

- **`C` is a world total, not a per-node ceiling. (decided)** Because `H ≤ 1`, `H^4 ≤ H^2` everywhere. With `C` as a literal per-node ceiling, an era transition from `k = 2` to `k = 4` would make every node except the peak lose population overnight: a node at `H = 0.5` would drop from `0.25·C` to `0.0625·C`. So the implemented form is

`Target_i = P_world * H_i^k / Σ_j H_j^k`

`P_world` is the era's world population: HYDE's historical total by default. Growth mechanics (see Growth drivers) can push it off history, but only the player's realm can push it *above* history. Bots' growth mechanics can only lower it (see Historical ceiling). The total is preserved by construction, `k` still funnels people into the hottest nodes, and the output can be checked directly against HYDE's totals. The ~150,000 ancient-megacity figure becomes a sanity check: at 300 BC with `k = 2`, the hottest node should come out around there. - **`k` must change gradually.** A step change in `k` at an era boundary redistributes the whole world in one tick. Interpolate `k` (and `C`) over the transition, the same way `H_hist` is interpolated between keyframes.

Historical ceiling: only the player exceeds history

Bots and unowned land can never hold more people in a place than real history had there at that date. Only the player can. (decided) Every node has a ceiling that tracks history as it happened, rises *and* falls:

```
Ceiling_i(year) = HYDE_pop_i(year)        (interpolated between snapshots, like
H_hist)
```

When real Babylon declines after Seleucia is founded, its ceiling declines with it, and a bot-held Babylon declines too. A bot can't keep it at its peak. Together with the `α` pull, which draws *under*-populated places up toward history, bot-held and unclaimed land hugs the historical record: never above it, and pulled back up when it falls below. (An earlier draft used the running historical maximum as the ceiling. That was wrong. It let a bot freeze Babylon at its peak forever, which contradicts the whole point of the historical pull.) Rules:

- **Applies to:** every node not owned by the player's realm. That covers bot realms, unorganized land held by bots, and unclaimed land.
- **Enforced on the population itself:** `Pop_i ← min(Pop_i, Allowance_i)` after each tick's update, where `Allowance_i = Ceiling_i` for any place the player never lifted above history. Capping only the *target* isn't enough. Population closes about 1.2% of its log gap to target each year, so when history falls it would lag far behind. Babylon's roughly 20x fall over three centuries would leave a bot-held Babylon about 2.4x above history the whole way down. Clamping the population makes bot-held places follow history's own decline at history's own pace. Targets are clamped too, so growth toward an unreachable target doesn't spill into the redistribution below.
- **Player legacy fades instead of snapping:** when a node the player held above its ceiling passes out of player ownership (conquest, or abandoning the land), `Allowance_i` starts at the node's population

at that moment and relaxes toward `Ceiling_i` on a log-scale curve of its own, `μ_legacy = 0.0115`: about 60 years to halve the excess on a log scale. It's kept apart from `μ`, which the growth fit set to 0.065; at that rate a lost city would be back to history within a generation. The city shrinks back to history gradually rather than overnight. Once `Allowance_i` reaches the ceiling, the node is an ordinary capped node again.

- **Capped excess stays with bots:** population trimmed off capped targets is redistributed to other *non-player* nodes still below their ceilings, in proportion to their targets. This keeps the world total on HYDE's historical figure. The player's own targets are never touched by it, so the player neither gains nor loses from bots hitting their caps. If every non-player node is at its ceiling, the excess is simply never born.
- **World total:** see `P_world` above. Only the player's realm can lift it above history.
- **Freedom below the ceiling:** bots still diverge from history *downward* and in how population shifts between places under their ceilings (wars, lost trade, moved capitals). They just can't create a metropolis where history never had one. Alternate-history *growth* is the player's alone.

A consequence worth knowing: a bot can't found a large new city in a place that was historically empty, since the ceiling there is near zero. Bots found cities where history did. The player is the only realm that can put a great city somewhere new.

Capital growth bonuses

Applies to **country capitals** (a realm's seat) and **province capitals** (each organized province's administrative seat), for both the player and bots. The ancient geographic regions (see Geographic regions) are purely geographic and have no capitals. For bots, the bonus only speeds growth *up to* the historical ceiling; it never lifts them past it. (DECIDED)

Two parts, both historically grounded, fitted to the targets above:

	One-time resettlement when designated	Ongoing driver bonus	→ top 50	→ top 10	Historical target
Country capital	**8%** of the largest city's population	**+0.05**	32y	76y	24 / 78
Province capital	**4%** of the largest city's population	none	60y	126y	62 / 122
Ordinary city	—	—	98y	165y	100 / 166

Fitted on the real HYDE grid with the base `β`, `μ` above (see Fit), measured on the regional engine. The country capital's top-10 time is 2 years early; its top 50 lands 8 years late, the closest the pair gets.

- **Resettlement** reproduces the founding decree. Seleucia was populated from Babylon, Samarra from Baghdad, and Antioch started with ~5,300 Athenian settlers from Antigonia plus their families. At 300 BC the largest node holds ~108,000, so 8% is ~8,600 people for a country capital and 4% is ~4,300

for a province capital, the size of a Roman veteran colony. The settlers are **moved, not created**: they're drawn from the realm's other nodes in proportion to their population. It happens only the first time a given city becomes that realm's capital of that kind, so toggling capitals can't farm settlers. For bots it's clamped to the capital's remaining headroom under the ceiling.

- **Driver bonus** stands for the court, bureaucracy, garrison, and tribute spent at the seat. It's added to the node's `drivers` in the `H_sim` feedback formula for as long as the city stays capital. When it stops being capital, the bonus goes away and the city declines along the normal `μ` curve, which is the Samarra and Pataliputra arc. On the real grid the fit wants it small (+0.05 for a country capital, none for a province capital): resettlement does nearly all the work.
- A faster growth rate (a `μ` multiplier) was tested as the bonus instead. It can't reproduce capitals' fast start: its best fit was 45 years to the top 50 against a target of 24. Resettlement is what produces the historical jump.
- **Province-capital targets are interpolated**, midway between country capitals and ordinary cities. A founding-to-rank dataset for provincial seats doesn't exist the way it does for world top 50 and top 10. Rough anchors support it: Basra and Kufa, garrison capitals founded 636/638, reached the top tier within ~30 years; Roman Carthage, a provincial capital refounded in 44 BC, took about 50 years to the top 50 and ~130 to the top 10; Fustat and Lugdunum took longer.

  (RETUNE WHEN BETTER PROVINCIAL DATA TURNS UP)
- **Capitals at the 300 BC start** already carry their bonus, and no resettlement is triggered for them. Starting drivers are backed out from HYDE so the historical start is still a fixed point with every existing capital's bonus included. Only capitals moved or founded after the start change anything.

Starting and seed populations

Everything starts from the historical population heatmaps (HYDE). (decided)

- **Game start:** every node's population is its HYDE population for 300 BC. The whole world begins as historical record, not procedural.
- **Seeding an empty node** (a founding on empty land, or the resettlement of a ruin) uses, in order: the node's HYDE population for the current year; else, for the player only, the median rural node density of its ancient geographic region. For bots and unclaimed land the seed is clamped to the ceiling. Where history had nobody that year, nobody but the player can settle. A new settlement starts at what history says that land held. Capital resettlement, above, adds to that seed.
- **Ruins need an event.** Log-scale growth never reaches zero on its own while a node has any target. So a settlement only becomes a ruin through an explicit depopulating event: razing, plague, or forced evacuation. A ruin stays empty until a realm deliberately re-founds it (reseeded immediately), or until a generation, **25 years**, passes and it's naturally resettled at its seed size. Resettlement doesn't restore the name. The ruin revives under its old name only when its cluster re-enters the naming percentile. Carthage was razed in 146 BC and refounded by decree in 44 BC. **(25 years is a starting value)**

Growth drivers: what makes people, and what moves them

(decided: approved as drafted, 25 Sep 2026; built, with every factor at its historical baseline: see As built, below)

Why this exists. The engine is calibrated for how fast population *responds* to `drivers`. Nothing yet defines what `drivers` are. Today every node's drivers are its historical heat and the gameplay slot (`driver_mods`) is zero, so the world simply follows HYDE. The one calibrated case is a single site given drivers of 1.0, the best site on the map. A stress test on the real 300 BC grid shows what an undefined driver formula would do. A contiguous, roughly Greece-sized realm (4,082 nodes, 2.3 million people) with every node at drivers 1.0 holds **21.0 million of the map's 39.5 million after 200 years**, and the largest city outside it falls from ~108,000 to ~24,000. At drivers 3.0 the rest of the map keeps 0.9 million. A single boosted city is harmless; broad, stacked drivers across a realm are not, and a player investing everywhere produces exactly that. The old toy-world constants break the same way, so this is structural, not a calibration problem. Three causes:

- **One map-wide pool.** `P_world` is fixed and heat is normalized by the map-wide maximum, so anything that raises one realm's heat lowers everyone else's. Attraction becomes a vacuum.
- **Nothing bounds drivers.** If factor effects simply add up, they stack without limit.
- **"Growth" and "movement" are the same number.** The engine has one heat term that decides both how many people there are and where they live, so it can't tell a baby boom from an exodus. The log-scale step also doesn't conserve people under strong drivers: moving every node a fixed share of its log gap doesn't preserve the sum, and the same stress test lost 2-4 million people that way, with nobody dying.

Principle: two channels, each bounded. (decided)

1. **Natural increase** (births minus deaths) decides *how many* people a region has. It's computed per ancient geographic region (see Geographic regions), from many factors, not just food.
2. **Attraction and migration** decides *where* they live. It conserves people exactly: moving someone never creates or destroys them.

`P_world` becomes the sum of the regional totals, replacing the single historical figure plus `player_world_delta` (removed).

Channel 1: natural increase, per region

```
P_r(t+1) = P_r(t) * (1 + g_r)
g_r = g_hist_r(year) + Σ_f w_f * s_f,r          (then bounded, below)
```

`g_hist_r` is the region's own historical growth rate, read from the HYDE keyframes. Each factor score `s_f,r` runs from -1 to +1 and is measured **against that region's historical baseline**: 0 means "as history had it". So a region nobody touches grows exactly as history did, and every factor is a deviation from the record, not an absolute rate the designers have to guess.

Real pre-modern growth was never one variable. Food mattered, but so did disease, violence, cities, burdens and luck. Factors:

Factor	What feeds it	Direction and shape	Historical grounding
Food security	Harvest (arable land x agriculture tech x irrigation x climate) against consumption, plus storage (granaries) and food imports along trade routes	Asymmetric: a deficit hurts far more than a surplus helps	Rome fed itself on Egyptian and African grain; famine years spiked mortality, good years barely moved fertility
Disease burden	Density, share of people in cities, sanitation (clean water, aqueducts, sewers), marshland and malaria zones, and trade connectivity, which spreads epidemics	Mostly negative; sanitation and medicine reduce it	Antonine Plague (165), Plague of Justinian (541), Black Death (1347) all travelled trade routes; the Pontine marshes and lower Mesopotamia were malarial
Urban penalty	Share of the region's people living in large nodes	Negative until sanitation and medicine tech offset it	Pre-modern cities had more deaths than births (the "urban graveyard"); they grew only by migration from the countryside
Security	War fought on the region's land, raids, banditry, occupation; peace and strong garrisons	War is strongly negative (burned harvests, flight, famine); long peace is mildly positive	The Pax Romana's growth; the third-century crisis and the Gothic wars in Italy
Land and living standards	Land per person against carrying capacity, wages, access to new land	Positive when land is plentiful (earlier marriage, more surviving children); falls to zero as the region nears capacity	Frontier and colonial populations grew fastest; crowded old regions stalled (the Malthusian trap)
Burden	Taxes and tribute, forced labor, slavery, serfdom, concentration of land into large estates	Negative, mild to moderate; heavy burden also pushes people to leave (channel 2)	Peasant flight from over-taxed land in the late Roman Empire
Health and public institutions	Medicine and public-health tech, famine relief, poor relief	Positive, era-gated; small in antiquity, large in modern eras	The mortality decline of the 1700s-1800s
Climate and harvest luck	A random yearly shock per region, not player-controlled; rare large events	Either sign; occasional severe negatives	The volcanic winter of 536; the Little Ice Age

Epidemics, famines and massacres are **events**, not factor scores. They remove a share of a region's people directly (up to ~35% for a Black Death-scale plague, spread over a few years) and are outside the rate bounds below. Their odds are what factors like disease burden and trade connectivity raise. Culture and family structure (for example later marriage) could become a small civ trait. (OPEN)

Carrying capacity. Every positive part of `g_r` is scaled by `(1 - P_r / Cap_r)`, where `Cap_r` is the region's arable land times the yield its agriculture tech allows (from the Resources density field). A region can approach its capacity but never grow past it; only better agriculture, irrigation or imported food raise it. That's the Malthusian ceiling that held every pre-modern region.

Bounds, anchored to HYDE. (decided; values retuned in balance tests) On this map, HYDE's century-averaged regional growth (5-degree regions) before 1700 runs from **-0.7% to +0.6% a year**, and the whole map from -0.2% to +0.23%. Only from 1700 do regions exceed +0.8%. So:

- **Factor deviation** `Σ w_f s_f,r` is clamped to **-2.0% to +0.5% a year** in the ancient and medieval eras. At +0.5% a year, a region can grow ~2.7x more than history over 200 years, a very strong alternate history but not a runaway. The upper bound rises by era with health tech.
- **Bots:** natural increase can only fall below history, never exceed it (only the negative part applies), which keeps the Historical ceiling rule. The player's regions get both signs.
- **Weights** `w_f`**:** a first proposal is to split the positive range across food 30%, land and living standards 25%, security 20%, health 15%, burden 10%. Disease, urban penalty and climate act on the negative side. These are starting values, to be set by the balance tests below. (OPEN)

Channel 2: attraction and migration (what `drivers` becomes)

`drivers` is redefined as **attractiveness**: how much a place pulls people from elsewhere. It's bounded and saturating:

```
A_i       = Σ_f v_f * a_f,i                          (pull minus push)
drivers_i = h_hist_i + (cap_era - h_hist_i) * sat(A_i)   if A_i ≥ 0
drivers_i = h_hist_i * (1 - sat(-A_i))                   if A_i < 0
sat(x)    = 1 - e^(-x)
```

With `cap_era = 1.0` in the ancient era, no place can become more attractive than the best site on the map, which is exactly the case the calibration measured. Stacking has diminishing returns: past a point, another road or market adds almost nothing. Pull and push factors:

- **Pull:** economic opportunity (trade routes, markets, ports, crafts, mines and other resource jobs), the seat of government (the existing capital bonus folds in here), security (walls, garrisons), public works (roads, water supply, harbors), open land for colonists (Greek colonies, Roman veteran colonies), and religious draw (sanctuaries, pilgrimage), a small one.
- **Push:** war and occupation, famine, epidemic, persecution, heavy taxes and tribute, loss of land to large estates.

Migration conserves people and is limited. (decided)

- **Within a region**, attraction redistributes the region's people the way the engine does now, heat to target to the log-scale step, but the region's total is renormalized after every step so it stays exactly `P_r`. That also closes the leak described above.

- **Between regions**, a net flow runs from regions whose average pull is below the map's toward those above it. It's capped at **0.2% of the source region's people a year**, and only between regions that share a border or a sea lane. Large historical streams (Greek colonization, the Germanic migrations) fit inside that over decades.
- **Forced movement** (deportation, resettlement of a conquered people, the Assyrian and Babylonian practice) is an explicit player action with its own limit, like the capital resettlement.

Prototype on the same stress test, with 5-degree blocks standing in for the regions until the region mask is built:

Greece-sized realm, maxed drivers, 200 years	Realm (2.3 M at start)	Rest of the map
Today: one map-wide pool	21.0 M	8.2 M
Regional totals	2.9 M	35.1 M
Regional totals + capped migration	3.0 M	35.0 M

The realm still gains, about 1.3x, but by drawing on its own region and a limited stream from its neighbours, not by emptying the map. Anything beyond that has to come from channel 1, which is bounded separately.

Calibration impact. The founded-city timings were measured with one map-wide pool. With regional totals, a new city draws only on its own region, so `β` and `μ` were refitted on the regional engine: `μ` is now 0.065 (see Fit). **(done)**

Balance tests: the contract

Each test runs on the real HYDE grid and must pass whenever a constant, weight or factor formula changes. They go into `tools/calibrate_population.py` and the headless test suite alongside the founded-city calibration. (decided; bounds retuned in balance tests)

Test	Setup	Must hold
Do nothing	Player realm, every factor at its historical baseline, 300 years	Tracks HYDE within ±5%
Maxed attraction	Greece-sized realm, every pull factor maxed, 200 years	Realm ≤ 1.5x its historical population from migration alone; no other region loses more than 10% to it
Maxed growth	Every natural-increase factor maxed, 200 years	Realm ≤ 2.7x history (the +0.5% a year bound) and never above its carrying capacity
Stacking	Double every factor input after saturation	Drivers change by ≤ 10%
Catastrophe	Worst war, famine and epidemic together for 10 years	Region loses ≤ 50%, and is back to 90% of its former size within 150 years of peace
Conservation	Migration only, natural increase off	Map total unchanged to within one person

Test	Setup	Must hold
Founded cities	The existing calibration	Timings stay near the historical targets

These bounds are what keeps a future economy, tech tree or resource system from breaking population: any system can raise a factor score, but none can push past what these tests allow.

As built (25 Sep 2026)

The groundwork is in: regional totals, conserving migration and the balance-test harness, with every factor at its historical baseline. No gameplay system writes a factor score yet; each will be wired in as it's built, and must pass the balance tests first. Code: `src/World/PopulationEngine.Growth.cs` (both channels) and `src/World/PopulationEngine.Regions.cs` (the region layer). A few details the draft left open were settled while building it:

- **History as the baseline, exactly.** A region's natural increase is HYDE's own year-on-year change for that region, times `(1 + deviation + recovery)`. With nothing changed, every region follows HYDE: over 300 years the worst region is within 0.12% of it.
- **Factor weights** are the proposal's: the +0.5% a year is split food 30%, land and living standards 25%, security 20%, health 15%, burden 10%. The -2% a year is split food 25%, security 25%, disease 20%, urban penalty 10%, burden 10%, land 5%, climate 5%. **(starting values, open)**
- **Mixed regions.** A region can hold the player and bots. Only the player's share of its people gets the positive part of the deviation. Bot-held nodes are still capped at history; growth that doesn't fit under their caps goes to the player's nodes in the region, but only as far as the region is above history.
- **Carrying capacity** is the region's food capacity from the crop model (see Crop yields and carrying capacity), never less than 5% above its historical population. Positive growth fades to zero as a region approaches it. (It was a 3x-history stand-in until the crop model.)
- **Recovery (new).** A region knocked below history regrows toward it at 1.5% of the log gap a year, never faster than +1% a year: empty land and plentiful food after a catastrophe, the land-and-living-standards factor's baseline. Without it a region halved by a plague would stay halved forever, since its historical rate only preserves the ratio.
- **Migration between regions** runs from a region toward neighbours with stronger pull (a shared border or a sea lane within 300 km). A region's pull is its people's average attraction. People keep moving until crowding balances the pull: at full pull a region settles at about 1.4x the people-to-history ratio of its neighbours (`MigrationPull` = 0.35). A region without the player can't take in more than its historical ceiling has room for.
- **Attraction and drivers.** Each node's attraction (`DriverMods`, pull minus push) saturates as drafted: `drivers = base + (1 - base) * sat(A)`. The capital bonus is added after that, so the calibrated capital timings still hold.
- **Whole people only.** HYDE gives many desert and steppe nodes less than one person. Those start empty, so regional history counts whole people only; otherwise sparse regions (Libya Interior,

Sarmatia) looked permanently under-populated, drew people in, and couldn't hold them. An empty node gets a share of its region's people only once it's due to resettle.

- **Events** (epidemic, famine, massacre) remove up to 35% of a region's people per call (`ApplyShock`), outside the rate bounds. Nothing triggers them yet.

Balance tests, first results (`src/Tests/PopulationBalanceTest.cs`, real grid, all pass; founded cities are `PopulationHydeTest`):

Test	Result	Limit
Do nothing (300 years)	Worst region 0.12% off HYDE, map total 0.008%	±5%
Maxed attraction (Hellas, 200 years, migration only)	Hellas 1.37x; hardest-hit neighbour (Phrygia) lost 3.0%	≤ 1.5x; ≤ 10%
Maxed growth (200 years)	Hellas 2.10x (its land feeds 5.9x); a bot region with the same factors 1.0000x	≤ 2.7x and capacity; bots never above history
Capacity binds (Syria, 400 years maxed)	Reaches 82% of the 6.3 million its land feeds, never more	≤ capacity
Stacking	Doubling a saturated attraction changes drivers by ≤ 5.0%	≤ 10%
Catastrophe (Italia: every factor at worst plus a 34% plague, 10 years)	Lost 39.4%; back to 90% 62 years after peace	≤ 50%; ≤ 150 years
Conservation (migration only, 100 years)	39.4 million held to within 6 people	one person per million (32-bit storage)
Founded cities	Ordinary 98/165, country capital 32/76, province capital 60/126	near 100/166, 24/78, 62/122

The conservation limit is one person per million rather than one person: each node stores its people as a 32-bit number, like HYDE and the saves, and adding 285,000 of them can't be exact to the person. The engine's own bookkeeping moves people exactly.

Build order. Channels and tests come before any system feeds them: regional totals, conserving migration, and the balance-test harness first, with every factor at its historical baseline (which must reproduce today's behavior). Then each gameplay system (resources, economy, tech, military) is wired to one or more factor scores and must pass the whole table before it merges.

3. Settlement naming and spawning (percentile-based)

- **Spawn:** a cluster becomes a named settlement when its population enters the **top 2-5%** of all active population clusters (population `> 0`; empty clusters are excluded so a large empty map doesn't shift the bar). This keeps the label count proportional to map size in every era.
  (DECIDED — 3% TO START; RETUNE WITHIN 2-5% IN PLAYTESTING)

- **Hysteresis:** once named, a settlement keeps its name, even when conquered or when its land becomes unclaimed, until it drops below the **top 20%**. The wide gap between entry (2-5%) and exit (20%) stops labels flickering on and off when a city hovers near the line. (DECIDED)
- **Ruins:** a named settlement whose population reaches exactly zero loses its active status but becomes a **ruin**: name and position are kept as a map marker, not deleted. It can be re-founded. Only depopulating events cause this; see Starting and seed populations. (DECIDED)
- **Scale shift:** at the planetary-to-interstellar transition the rule doesn't change, only what a "cluster" is. The top 2% of hexes becomes the top 2% of hemispheres or core worlds. (DECIDED)

Useful property: **the ordering of target populations depends only on the ordering of `H_final`, not on `C` or `k`.** A percentile rank is unchanged by any increasing transform, and `C * H^k` is increasing in `H`. So retuning `C` and `k` changes displayed populations but never the settlement ranking the world is heading toward. Only the heatmap changes that. (Actual populations lag their targets by the inertia above, so the names on the map settle into that ranking over decades, not instantly.) Calibration and map readability can be tuned independently.

Cluster vs. node vs. cell. (decided) The percentile has to rank *clusters*, not grid cells. The ownership grid is 8192x5476 at ~0.59 km² per cell, so there are millions of land cells, and the top 2% of *cells* would be hundreds of thousands of labels. A single city also covers many adjacent cells. So:

- **Node:** the unit `H` and `Pop` are computed on. Coarser than a cell, roughly HYDE's native resolution (5 arcminutes, ~9 km), since there's no finer truth to compute against. Per-cell density values are sampled from their node.
- **Cluster:** a local maximum of `H_final` plus the contiguous nodes that drain to it (a watershed on the heatmap). Its population is the sum over those nodes. Nearby peaks closer than a minimum separation merge, so twin peaks don't produce two labels for one city.

Global ranking with a regional floor. (decided) A purely global percentile would put almost every 300 BC label in the Aegean, the Nile, Syria, and Mesopotamia. That's historically honest, but it leaves regions like Libya Interior or Scythia with no labels at all. So the ranking is global, plus a regional floor: any region with no named settlement gets its single largest cluster named. Hysteresis and ruin rules apply to floor-named settlements too. Regions are a fixed geographic partition, not political borders, so conquest never changes which floor applies.

Geographic regions — the ancient geographers' map (built)

The partition uses the regions the Greco-Roman geographers themselves drew (Hecataeus, Herodotus, Eratosthenes, Strabo, Ptolemy), grouped under their three continents: **Europa**, **Libya** (their name for Africa), and **Asia**. The map's extent (longitude -10 to 55, latitude 10 to 48) is almost exactly the world those geographers described, so their divisions cover it without inventing names. Continent borders follow the ancient convention: the Tanais (Don) between Europa and Asia, the Nile/Isthmus line between Libya and Asia. (DECIDED)

Continent	Regions
Europa	Hispania, Gallia, Raetia, Italia (with Sardinia and Corsica), Sicilia, Pannonia, Illyricum, Dacia, Thracia, Macedonia, Hellas (with Crete and the Aegean islands), Scythia
Libya	Mauretania, Numidia, Africa (the Carthaginian heartland, the name the continent itself later took), Syrtica, Cyrenaica, Marmarica, Aegyptus, Libya Interior (the Sahara), Aethiopia (everything south of Egypt, Kush included)
Asia	Lydia, Phrygia, Cilicia (with Cyprus), Cappadocia (with Pontus), Colchis (the Caucasus), Armenia, Syria (with Phoenicia and Judaea), Mesopotamia (with Assyria), Babylonia, Arabia Petraea, Arabia Deserta, Arabia Felix, Media, Susiana, Persis, Hyrcania, Sarmatia

That makes 38 regions. Smaller neighbours are folded into the larger unit named above rather than given their own entry. As with Europa, several of these names are the ancient roots of modern ones (Africa, Arabia, Armenia, Syria, Libya).

As built (25 Sep 2026). `tools/build_region_mask.py` bakes a region id for every population node into `data/regions/` (`region_nodes.u8.gz` and `regions.json`, which lists each region's continent, what's folded into it, where its name goes, and its neighbours). Each region is a hand-drawn outline in longitude/latitude whose borders follow the ancient descriptions snapped to physical features: the Pyrenees, the Alpine crest, the Danube and Sava, the Nestos, the Tanais (Don), the Caucasus, the Taurus, the Euphrates, the Zagros, and the Nile/Isthmus line. The coastline comes from the land data, not the outlines. Islands go whole to their region (Lesbos, Rhodes and the other east Aegean islands to Hellas although they lie off Asia's coast; Cyprus to Cilicia; Sardinia and Corsica to Italia). What each region takes in, where the table above doesn't say:

- Raetia takes Noricum and the strip north of the upper Rhine (the map's only bit of Germania); Gallia takes the Helvetii.
- Illyricum takes Dalmatia, Dardania and Upper Moesia; Thracia takes Lower Moesia and the Dobruja; Dacia runs to the Tyras (Dniester) and takes the Tisza plain; Hellas takes Epirus.
- Lydia takes Ionia, Aeolis, Mysia, the Troad, Caria and Lycia; Phrygia takes Bithynia, Galatia, Pisidia and Lycaonia; Cilicia takes Pamphylia; Cappadocia takes Paphlagonia; Colchis is the whole Caucasus (Colchis, Iberia, Albania).
- Hyrcania takes Parthia and the Dahae east of the Caspian; Sarmatia is everything beyond the Tanais north of the Caucasus and the Caspian; Persis takes Carmania.
- Aethiopia is everything south of Egypt, and the Sahel south of the 16th parallel; Libya Interior is the Sahara between.

Two regions are neighbours if their land touches or a sea lane joins them (coasts within 300 km across water); migration only runs between neighbours. Borders are exact to the population grid (~9 km), which is the finest data the regions are used on. **(boundaries can be redrawn in the tool and re-baked; regions beyond the current extent, if the map grows, get names the same way, from the geographers who described those lands)**

Display: only on the "Regions" map overlay. (decided, built) Region names never appear on the default map view. They show only when the player switches on the **Regions** overlay (the Regions button, or the R key), which tints each region in its continent's colours (Europa blues, Libya ochres, Asia reds), draws a faint boundary between regions and their names, and hides realm borders and names while it's on. The province panel also names the region a province lies in. The regional-floor rule runs all the time regardless of which overlay is showing.

Resources and the land (decided, 25 Sep 2026; land layer built)

In short

What every place is (terrain, climate, soil, water, 300 BC forests and farms, mines and special goods), baked from real scientific data that allows the game to be sold, with hand fixes where it's wrong for 300 BC. Each crop, tree and herd has its own needs, and from them the game knows how many people every place could feed; that limits how far a region can grow. You can see all of it on the map.

The designer's resource plan, with suggestions folded in; the designer's answers are in Design decisions. Step 1, the land layer, is built (see The land layer as built, below).

Three tiers, not one list

1. **The land** (what a place *is*): soil, water, climate, terrain, living habitat, geology. Stored per population node (~9 km, 284,626 land nodes), the same grid as population and regions. That's fine enough for fields, forests and ore bodies and small enough to simulate: 40 numbers per node is ~45 MB. The 45-million-cell ownership grid is too fine to simulate and has no finer data behind it.
2. **Raw resources** (what a place *can yield*): crops, orchards, herds, fish, timber, stone, clay, ores, salt. Each is a *potential* per node, computed from the land layer by a suitability profile, times how much of it is actually worked (people, tools, tech).
3. **Goods** (what people *make*): flour, bread, beer, olive oil, wine, yarn, cloth, charcoal, bricks, pottery, bronze, iron, tools, weapons, ships, and the prestige goods. These aren't on the map at all: they're **recipes** (inputs, labour, fuel, a workshop or skill) run where the inputs and the people are. That keeps the map data small and makes the full goods list cheap to add to.

Provinces and realms sum all three, as they already do for population (see Layers: stats are always summed, never stored on the province).

The land layer

The designer's six parts, each a handful of numbers per node:

Part	Numbers per node	Drives
Soil	depth, texture, drainage, organic matter, N/P/K reserve, salinity, acidity, erosion, compaction	which crops thrive, yields, how fast farming wears a field out, whether drainage or manure helps
Water	seasonal rainfall, river flow, floodplain moisture, springs, groundwater depth, water quality, irrigation supply	crop survival, herd size, gardens and orchards, settlement capacity, mills, river transport

Part	Numbers per node	Drives
Climate	growing season, heat, frost, drought frequency, prevailing winds, storm exposure	crop suitability, harvest reliability, animal survival, sailing seasons
Terrain	slope, elevation, aspect, flood risk, passability	usable farmland, terracing cost, grazing, roads, mines, transport
Living habitat	shares of grassland, woodland (with species mix and age), scrub, wetland, river habitat, coastal nursery grounds	forage, timber, reeds, honey, medicinal plants, fish and shellfish
Geology	ore bodies (grade, size, depth), workable stone, clay beds by use, salt, sand, minerals	mining, building, metalworking, pottery, glass, pigments

Suggestion: bake it from real data, the way HYDE gives population, so the map is history, not invention. Candidates: soils from HWSD/SoilGrids; climate from WorldClim or CHELSA (with CHELSA's palaeoclimate for 300 BC); elevation and slope from the relief data the terrain already uses; rivers from HydroSHEDS; 300 BC forest cover from the KK10 reconstruction (which models ancient deforestation); ores from the USGS mineral database plus the archaeological atlases of ancient mines and Roman quarries. HYDE itself has cropland and pasture grids for 300 BC: a direct check that the farming model puts fields where history had them. Each becomes a `tools/build_*.py` step, like the region mask.

Each crop and herd its own profile

As the designer's additions say: no generic "farmland". Each crop, tree and animal has a suitability profile over the land numbers (e.g. olive: no hard frost, deep well-drained soil, tolerates drought and slope; date: heat plus a reachable water table even under a desert sky; emmer tolerates poorer soil than bread wheat). A node's yield for a crop = best-case yield x suitability x the share of land worked, lowered by nutrient depletion, raised by manure, fallow, irrigation and tools. Herds eat forage, and overgrazing lowers next year's capacity; dung goes to fields *or* to the fire, a real choice.

Change over time: four speeds

The designer's three, plus one:

- **Fixed** on game timescales: bedrock, slope, ore body locations, the shape of a natural harbour.
- **Renewable but vulnerable**: soil nutrients, standing timber, forage, fish stocks, groundwater. Each has a stock and a regrowth rate, and can be run down.
- **Seasonal**: rainfall, river level, soil moisture, grass, navigability. Suggestion: with years (or more) per turn, seasons aren't simulated one by one; each year draws its weather from the node's climate (drought frequency, flood risk), so harvests vary and a run of bad years is possible. Sailing seasons become a cap on how much sea trade a year can carry.
- **Slowly altered by people** (added): deforestation (Lebanon's cedars, the Greek hills), salinisation from irrigation (southern Mesopotamia), harbours silting up (Ephesus, Miletus, Ostia), marshes drained or spreading. These change the "fixed" layer over centuries.

History as the guide, here too

Like population's historical ceiling, bots' land and extraction should follow history: Laurion's silver fading, Spanish mines booming under Rome, Mesopotamia's soils salting, forests shrinking where they did. The player alone can do better (or worse) than history. (DECIDED) Each historical site carries the years it was worked (Laurion to about 100 BC, Las Médulas from about 20 BC), ready for that.

Goods list: additions and grouping

Suggested additions, all historically important on this map in 300 BC: sesame and sesame oil (Mesopotamia's oil, not olive), fodder crops (vetch, lucerne), cotton (a little, Egypt and beyond), natron (Egypt: glass, washing, embalming), bitumen (Dead Sea and Mesopotamia: caulking, mortar), alum (dyeing), cinnabar (red pigment, mercury), silphium (Cyrene's famous and later extinct export: a perfect vulnerable resource), cedar as its own ship and temple timber, garum (fish sauce), leather, war elephants (North African and Syrian herds, captured, not bred), ivory as a raw good, and goods from beyond the map edge (amber, silk, pepper and spices, Indian steel) arriving only by trade.

Suggestion: keep the designer's full list as the *goods*, but show and simulate them through ~12 **families** (the designer's own headings: field crops, orchards, animals, fish and gathering, grain products, oil and drink, fibre and cloth, wood and fuel, stone and earth, metals, metalwork, construction, prestige). The engine tracks every good; the screens lead with families and open to detail. Timber should have a few grades (fuel, building, long straight ship timber), since the designer notes that woodland good for fuel isn't ship timber.

Labour: in antiquity much work was done by enslaved people. Suggestion: model labour as a population input (free, dependent, enslaved), not as a tradeable good. **(decided, with the designer's note: history is king. Labour comes from the population, split as history had it; the slave trade exists as it did, and slavery is abolished as history abolished it.)**

Goods and recipes (built, 25 Sep 2026)

`data/goods.json` holds 148 goods in the designer's 13 families. Amounts are **loads**: what one person makes in a year of full work on ideal land.

- **Raw goods** come from the land layer: each crop and herd from its suitability map, metals, stone and prestige goods from the resource maps, logs from woodland, reeds from wetland. Country people spend a fixed share of their work in each kind of production (fields 42%, herds 16%, orchards 10%, gathering 5%, forests 6%, mines 3%, clay pits 3%), spread over what their place yields, favouring what grows best there.
- **By-products** come with their source: wool and cheese with sheep, hides, horn, tallow and dung with cattle, straw with wheat, beeswax with honey.
- **Made goods** follow recipes, in order (inputs first): grain is threshed, milled and baked; grapes and amphorae make wine; ore and charcoal make bloom iron, then wrought iron, tools and weapons; copper and tin make bronze. Some recipes take any one of their inputs (threshed grain from wheat,

barley, emmer or rye; fuel from wood, brushwood, dung, straw or reeds, so dung burned is dung not spread). How much is made follows what people need and what craftsmen sell, passed back up each chain; people's needs come first, and inputs are shared fairly among the recipes that use them.

- **Needs**: each person uses bread, grain, pulses, vegetables, milk, cheese, fish, salt, fuel, pottery, oil, wine, cloth and tools each year. What a realm can't make shows as a shortage (trade will cover it, section 3).
- **Value and taxes**: every good has a price in drachmae (a made good's is its inputs plus its labour at 200 drachmae a year). The realm's output is the value of all it makes, less what was used up making it, plus the townspeople's trade and services. Taxes and tribute are a share of this output (it replaces the flat 100 drachmae a person).
- **On screen**: the Goods button (or G key) opens the panel: value by family, each opening to its goods (made, used, needed, spare or short).

Harvests, weather and renewable stocks (built, 25 Sep 2026)

- **Weather** (decided: a yearly draw). Each geographic region draws its harvest every year: 1 is an ordinary year. The draw mixes a shared part (a bad year across the whole map) with the region's own, and swings more where rain is unreliable (the land layer's rain variability) and less where rivers and irrigation water the fields. The same year always has the same weather. Below 75% is a famine, above 120% a bumper year; both make the chronicle in the player's lands.
- **Stocks** per region, each 100% as nature left it: soil nutrients, standing forest, pasture forage and fish. They wear down when the region's people press on what its land can feed (above half its carrying capacity) and recover when rested: fish within a few years, pasture in a decade or two, soil in decades, forests in a century. Dung spread on the fields speeds the soil's recovery; dung burned as fuel doesn't (the designer's trade-off).
- **Effects**: field crops yield harvest x (50% + 50% x soil); orchards feel the weather less; herds follow pasture; fishing follows fish; timber follows forest. Carrying capacity follows soil and pasture. The growth drivers' Climate factor takes the harvest and the Food factor the state of the land (bounded, so history stays on course).
- **On screen**: the province panel shows its region's harvest this year and its soil, forests, pasture and fish.

Trade (built, 25 Sep 2026)

- **Who trades**: realms linked by a border or a trade route, and through middlemen (see "Trade routes" below; this replaced the first rule, where every coastal realm traded with every other). Realms at war don't trade.
- **What**: once a year each realm offers what it can spare (after its own people and crafts) and asks for what its people lack. A seller shares its spare goods among its partners in proportion to what they lack.

- **Prices**: one price per good, its usual price times the square root of want over supply, between half and three times.
- **Money**: buyers pay and sellers earn at that price. Merchants keep 10% of what they carry (added to the realm's output), and the state takes 5% in tolls and customs, a new line in the treasury.
- **Needs and unrest**: the share of people's needs met (by value) is shown in the Goods panel. Below 80%, want adds unrest in every province of the realm (up to 40% at nothing).
- **From beyond the map**: amber (by the Amber Road, arriving in Pannonia and Scythia), silk (Hyrcania, Media), pepper and spices (Arabia Felix, Persis, Aethiopia) and Indian steel (Persis, Hyrcania) arrive each year in whoever holds those regions, and travel on only by trade. They are wanted, not needed.
- **Farming richness**: where the fields yield more per hectare than the typical farmer's (the watered Nile valley), each farmer grows more (up to 2.5x), so Egypt sells grain.
- **Slaves**: the slave trade comes with labour (section 4), since people are counted by the population, not as goods.

Labour and people (built, 25 Sep 2026)

Decided: history is king. `data/labour.json` gives, for every geographic region, the shares of its people who were **dependent** (bound tenants, serfs, Egypt's royal farmers, helots, clients) and **enslaved**, at points through time (Italy's enslaved share grows from about 10% in 300 BC to a quarter in the late Republic, then falls as tenancy and serfdom rise), interpolated between them, and the year slavery was abolished in law in each region's lands (from then the enslaved share is 0). The rest are free. The figures are approximate (Scheidel 2008; the Cambridge Economic History of the Greco-Roman World) and meant to be refined.

- **Levies**: only free men serve in full; dependents give half as many; the enslaved none. A realm's manpower is scaled by this.
- **Mines**: the enslaved worked the mines (Laurion, the Spanish silver mines): a region's mine output is multiplied by 1 + 2 x its enslaved share.
- **Unrest**: where more than 15% are enslaved, unrest rises (the risk of servile war).
- **Captives and the slave trade**: a conqueror carries off 5% of a taken province's people as captives (not after abolition there). Captives count among the realm's enslaved; 5% a year die or are freed; beyond 5% of the realm's people, they are sold in the slave markets at 300 drachmae each, and the silver reaches the treasury.
- **On screen**: the Treasury and Army panel shows the realm's free, dependent and enslaved shares and its captives, and the sale of captives in the accounts.

What it feeds

- **Carrying capacity**: the food a region's land can grow: calories from crops, herds and fish at its current tech (built; see Crop yields and carrying capacity).

- **Growth factors**: food security (harvest against need), land and living standards (land per person), disease (marshes, malaria zones).
- **Attraction**: mines, ports, markets and quarries pull people.
- **Economy, military and Might** (later build steps): output, what can be built, which units can be raised.

Suggested build order within step 2

1. The land layer: bake the six parts per node from real data, show each as a map view, and check it against known history (fields where HYDE has cropland, forests where KK10 has them).
2. Raw-resource potentials from the suitability profiles, and carrying capacity from food, replacing the 3x stand-in. Re-run the population balance tests.
3. Renewable stocks (soil nutrients, forests, forage, fish) with depletion and regrowth, and the yearly weather draw.
4. Goods and recipes, with the economy (build step 4), where they're first needed.

Each part gets balance tests like population's: a realm doing nothing tracks history, maxed effort stays within bounds, a depleted stock can recover.

The land layer as built (25 Sep 2026)

83 values for every one of the 284,626 land nodes, in `data/land/` (6.4 MB), baked by `tools/build_land_layer.py` from sources fetched by `tools/fetch_land_sources.py`, plus `tools/land_sites.json` (88 historical sites and the hand fixes). Every source allows selling the game:

Part	What's in it	Source (licence)
Terrain	elevation, steepness, ploughable flat share, ruggedness, which way slopes face, distance to sea, coastal shallows	ETOPO 2022, NOAA (public domain)
Climate	average, summer-high and winter-low temperature, rainfall and its seasonality, growing season, frost days, dryness, wind, storms, plant growth; derived drought risk and year-to-year rain variability	CHELSA V2.1, 1981-2010 (CC0)
Soil	clay, sand, silt, organic matter, nitrogen, pH, nutrient holding, stones; derived depth, drainage, fertility, salinity, erosion, compaction	SoilGrids 2.0, ISRIC (CC BY 4.0)
Water	largest river, distance to a river, lakes; derived floodplain, marsh, groundwater, irrigation water, springs, water quality	Natural Earth (public domain)
Habitat, 300 BC	farmed and grazed, forest (and its conifer share), scrub, grassland, marsh, desert; shares add up to one	KK10, Kaplan et al. (CC BY 3.0), with natural cover derived from climate
Resources	39 goods: metals, salt, sulfur, bitumen, natron, marble and other stone, fine clay, glass sand, cedar, ship timber, papyrus, murex, frankincense, myrrh, balsam, silphium, horses, elephants, ivory…	USGS MRDS (public domain) and the historical sites

Choices made while building it:

- **Modern climate, not ancient.** The ancient-climate version of CHELSA (TraCE21k) is licensed CC BY-SA: "share-alike" would bind the game's own data files to that licence. The CC0 modern averages are used instead; the Roman Warm Period was close to them, and known differences are hand fixes.
- **Hand fixes where modern data is wrong for 300 BC:** KK10 models farming from rainfall, so it put Egypt's Nile valley at ~1% farmed; the fixes set the Nile valley, the delta, the Faiyum and Babylonia as farmland and floodplain. Also the Mesopotamian marshes, the Pontine marshes, Lake Copais, Lake Moeris and the delta lagoons (all drained or shrunk since), and the salting of southern Mesopotamia.
- **Growing season** is CHELSA's rain-fed one, so irrigated Egypt shows a short season: irrigation extends it, which the crop model will use.
- **Deposits** from USGS are modern knowledge, counted at half strength as potential; the historical sites are what antiquity actually worked, each with its years.

Map views show it in the game: the Map view menu (bottom right) lists Political, Regions and 52 land views grouped by part; a legend top-left explains the view, names its source, and shows the value under the mouse. `src/Tests/LandLayerTest.cs` checks known places (the Nile floodplain, Alpine frost, Laurion's silver, Tyre's murex…), that habitat shares add up to one, that farmland follows where HYDE puts people (r = 0.61), and that every source's licence is cleared for a commercial game.

Next: raw-resource potentials from per-crop suitability profiles, and carrying capacity from food, replacing the growth drivers' 3x stand-in (built: see below).

Crop yields and carrying capacity (built, 25 Sep 2026)

`src/World/CropModel.cs` reads 31 profiles from `data/crops.json` - 13 field crops (wheat, emmer, barley, millet, rye, oats, lentils, chickpeas, peas, broad beans, sesame, flax, hemp), 8 orchards and gardens (olives, grapes, figs, dates, pomegranates, almonds, walnuts, vegetables), 9 herds (sheep, goats, cattle, pigs, horses, donkeys and mules, camels, poultry, bees) and fishing - and scores every place 0-1 for each from the land layer. Each profile has its own needs, as the designer asked: rain (or irrigation) in a range, temperature, the winter cold it survives, the summer heat it needs to ripen, pH, salt tolerance, drainage, soil depth, and how much it depends on fertility (legumes make their own nitrogen). Some needs are particular: dates need a water table or irrigation whatever the rain; millet and sesame are summer crops, so north of the Sahara's dry summers they need irrigation; winter-sown grain is judged on the cool season it grows in (Egypt's winter wheat and barley); grapes like south-facing slopes; goats like rugged ground; cattle need water. Yields are ancient ones on good land (wheat 1.1 t/ha, barley 1.1, emmer 0.9, olive oil 0.25, dates 3…). The numbers are data, so they can be adjusted without touching code.

Carrying capacity of a place = the food it could grow at most with ancient methods, in people fed (2,000 kcal a day, less 15% lost):

- **Fields**: the best food crop on the flat land. Rain-fed farming sows at most 40% of the flat land (the rest is woodland for fuel and pasture for the plough oxen) and lies fallow every other year; irrigated land is sown up to 85% and cropped yearly.

- **Orchards and terraces**: the best orchard on 20% of the slopes.
- **Herds**: milk and meat from the rest, by how much it grows and what it is (grass, scrub, forest, fallow, marsh, desert).
- **Fish**: coastal shallows, rivers, lakes and marshes.
- **The ancient plough**: yields on heavy clay are up to 35% lower: the ard scratches light soils, and heavy clay waits for the mouldboard plough, a technology to unlock later (the Po valley stays modest until it's drained and ploughed deep, as happened).

It computes in about a second at load. Against history (HYDE), in people:

Region	Food capacity	300 BC	Peak before 1700
Italia	18.2 M	4.4 M	12.6 M (69%)
Aegyptus	11.7 M	3.2 M	4.9 M (42%)
Syria	6.3 M	2.4 M	5.9 M (94%)
Gallia	29.2 M	2.4 M	14.2 M (48%)
Hellas	7.3 M	1.2 M	1.5 M (21%)
Sarmatia	19.8 M	0.1 M	0.8 M (4%)
Whole map	479 M	39.4 M	

Every region held fewer people in 300 BC than its land could feed; long-settled farming lands came close to their capacity by 1700, while the steppes and frontiers stayed far below it, as history had them.

It replaces the growth drivers' stand-in. A region's carrying capacity is now its food capacity - but never less than 5% above what history actually had there, since history proves that food existed (grown, herded or imported). Positive growth fades to zero as a region nears it. A new balance test checks it binds: a player-held Syria with every growth factor maxed for 400 years reaches 82% of the 6.3 million its land can feed, and never more.

Map views: a "Crops and food" group - food capacity (people per km²) and each crop, tree, herd and fishery's suitability. `src/Tests/CropModelTest.cs` checks crops grow where history grew them (olives at Athens, not in the Alps; dates on the Nile, not in Greece; camels in Arabia; millet in the Sahel) and the capacities against HYDE.

Still to come in this step: yearly harvests from the weather draw (the land's drought risk), and the growth factors' food security read from harvest against need.

Cultures and loyalty (built, 25 Sep 2026)

Every province has a people (culture) and a religion, as in 300 BC (`data/cultures.json`: a default for each geographic region, with provinces that differed, such as Latin Latium, Etruscan Etruria, Jewish Judaea, Punic western Sicily). Each playable realm has its ruling culture (the Ptolemies and Seleucids are Macedonian); kin cultures (Latin and Italic, Greek and Macedonian, Punic and Phoenician, ...) count as close.

- **Integration** (0 to 100%): how fully a province is part of the realm. Provinces held at the start begin at 100% if they share the ruler's culture, 80% otherwise. A conquered province starts at 0% (50% if it is the conqueror's own people) and rises each year: fully in about 10 years among one's own people, 40 among kin, 100 among foreigners; half as fast while the realm is at war or taxing heavily.
- **What it changes:** a province pays tax and gives levies at 50% + 50% x integration; the rest of its people count as unorganized (tribute only).
- **Unrest:** 0.6 x (1 - integration), plus tax (-10% low, +15% heavy, +35% crushing), +10% for a different religion, +30% for each 100% the realm's province count exceeds the court's capacity of 15.
- **Revolt:** only in the player's realm (other realms' revolts follow history, decided). Above 50% unrest, a province has a yearly chance of (unrest - 50%) x 0.3 to rise and break away as unclaimed land.

Score and goals (built, 25 Sep 2026)

Decided: a sandbox with optional goals. `data/goals.json` gives every playable realm history's ambitions for it (Rome: unite Italy, take Sicily, break Carthage in Africa, rule "our sea"; Egypt: hold Coele-Syria, keep Cyrene, rule the Aegean, the riches of the Ptolemies; and so on) plus common goals (double your people, rule 30 provinces, 10,000 talents and no debt, still standing in AD 1). A region goal needs 60% of the region's people in your land. Reaching a goal is checked every year, marked with the year, and makes the chronicle.

Score = people / 100,000 + 1 a province + (treasury - debt) / 1,000 + the points of every goal reached. The Goals button in the top bar shows the score; its panel lists each goal with how near you are.

First-game guide: a "How to play" panel opens the first time the game is played (and from the Goals panel any time).

Balance fixes with it: - Rome's Italian allies paid their own contingents, so Rome pays half its army's upkeep (start_realms.json, upkeep_share 0.5). With it the AI's Rome fights the Pyrrhic War (281 BC), both Punic Wars (264 and 218 BC) and Cisalpine Gaul (225 BC) on history's timetable. - Other realms no longer hoard: with three years' income saved they lower their taxes and spend a fifth of the excess each year on temples, palaces and largesse; at war they spend their war chest over about ten years; in debt they raise taxes. - Random draws (weather, battles, revolts) now come from a hash that is the same on every run, so a saved game and the tests always play out the same way.

Armies and sieges (built, 26 Sep 2026)

Decisions "Playable 9, 10, 12": named armies based in provinces, every culture's own units, sieges over years.

- **Units** (`data/units.json`, 52 kinds): every people's own, in seven roles (heavy and light infantry, archers and slingers, cavalry, horse archers, elephants, warships), each with its Might, men, cost, upkeep, a resource it needs, and a description: hastati and principes, Companion cavalry, the Sacred Band, Numidian horse, scutarii, warbands, falxmen, Scythian horse archers, cataphracts, Kushite bowmen, camel riders… A realm raises the units of its own people and of every people whose province it holds (Carthage, holding Numidians, raises Numidian horse), plus anyone's levies, skirmishers, slingers, light horse, triremes and (with elephant country) elephants.
- **Armies**: each realm has named armies (Rome's are Legio I, II…; the Hellenistic kings' the Royal army; the tribes' warhosts), each standing on a place on the map with its own list of units and its own weariness. The Armies panel raises units into an army, sends units home, moves an army anywhere in your own land (click Move, then the map), founds new armies at the capital, and chooses the army Conquest mode paints for. Armies of yours and of your enemies show on the map.
- **Reach** is now per army: 400 km over land from where it stands in a year; with warships in the army the sea costs a third as much.
- **Sieges**: painting enemy land and ending the turn sends the chosen army to lay siege; it marches to the place. Each year a siege advances by 0.35 x (besiegers / garrison)^0.6 x (realm Might ratio, clamped 0.5-2)^0.3 (between 5% and 100% a year): about three years between equals, one year for an overwhelming army. The garrison is the place's militia, the province's walls (4 Might) and any of the owner's armies standing there. Each year the defender may march a relief army (60% of its armies not besieging, shared among its sieges) against each siege: a battle by the usual odds; if the besiegers lose, the siege is lifted. Besiegers lose 2% a year to disease and desertion and tire. When a siege is complete the place changes hands, captives are taken and the war score moves. An army split between several sieges divides its strength.
- **Other realms** besiege the same way with their main army, a few places at a time.

Pretexts, war exhaustion, peace terms, alliances and vassals (built, 26 Sep 2026)

Decisions "Playable 15, 16, 17, 19".

- **Pretexts** (in the spirit of Stellaris' casus belli, not a copy): historical claim (history's own war for your realm in its years), reconquest (land you held within 50 years), free our people (they rule a province of your culture), border dispute (neighbours), subjugation (a neighbour with a third of your Might or less: the prize may be their submission), war of faith (neighbours with other gods), or none at all. Each costs aggression on declaring (0 to 25) and for each province taken (1 to 8).
- **Aggression** (0-100) fades by 2 a year. It lowers every realm's opinion of you (0.8 a point). At 50 or more, neighbours who think ill of you (-20 or worse) form a coalition: they defend each other against you, and a coalition half again stronger than you may strike first.
- **Opinion** of another realm, with its reasons shown: kindred people (+10/+20), the same gods (+10) or others (-5), allies (+40), overlord and vassal (+20), at war (-60) or a recent war (-20), old rivals (-30: Rome and Carthage, the Ptolemies and Seleucids, the Antigonids and the other Successors, Carthage and the Sicilian Greeks, Rome and the Gauls), a marriage between their ruling houses (+15), and fear of your conquests.
- **Alliances**: each defends the other. When a realm is attacked, its allies, coalition partners, vassals and overlord join (a far-off ally of another realm stays out, as history would; the player's allies and the allies of a realm the player attacks always come). In force in 300 BC: Seleucus with Lysimachus, Ptolemy with Lysimachus. A realm accepts your alliance at an opinion of +25; friends facing your enemies offer theirs. Breaking one costs 5 aggression.
- **Vassals** pay their overlord a tenth of their income a year, fight in its wars and make none of their own. A far weaker realm (a third of your Might) that thinks well of you may submit without a war; or vassalage is a peace term.
- **War exhaustion** (0-100 each side): +3 a year, +30 for losing a realm's sustainable manpower in men, +8 for each province lost, +2 for each of its places under siege. At 100 a side must make peace: the winner takes the places it has half besieged, and tribute if it leads by 40.
- **Peace terms by war score**: white peace; each province (8-40 by its share of the loser's people; half if already half besieged, half again if it was once yours); tribute (20); vassalage (60, on a subjugation pretext or with 60 war score). The loser pays up to its losses plus half its exhaustion. The Diplomacy panel shows each term's cost and whether they would accept.
- **Appeasement** (only toward the player, or as history allows): a beaten or exhausted enemy sues for peace offering silver and the places already falling; a city under siege may offer a ransom (about 20 drachmae a head) to have the siege lifted. Offers wait in the Diplomacy panel for two years.
- **Other realms** declare history's wars only with half the defender's strength (its allies counted), concentrate their army on one new siege a year, and press where they can reach when history's regions are out of reach.

Buildings, province tabs, province taxes and remedies (built, 26 Sep 2026)

Decisions "Playable 5, 6, 31, 32".

- **Buildings** (`data/buildings.json`, the core set): irrigation works (+25% field crops; needs a river or floodplain), granary (a bad harvest hurts half as much), terraces (+30% orchards; needs hills), mines (+50% ore and stone; needs a deposit), workshops (+15% made goods, weighted by the province's share of the realm's people), market and fora (+10% trade profits and customs, likewise), harbour (+30% fishing, +10% trade, warships a quarter cheaper for the realm; needs a coast), roads (+25% integration speed, +5% trade), city walls (+4 garrison Might, can be built twice), barracks (+20% levies from the province), great temple (-10% unrest, +20% integration speed), aqueduct and baths (-5% unrest, better health for the region's growth; needs a town). Each costs silver, takes years (halted while besieged) and has upkeep (added to administration). One building at a time per province. Other realms with silver to spare and at peace build too.
- **Province panel** (essentials plus detail tabs): name, realm, area, people and region at the top; tabs for People (culture, worship, who works, integration, unrest, the province's own tax rate), Land (harvest, soil, forests, pasture, fish, resources), Buildings (built, under way, what can be built with costs and reasons) and Military (sieges, armies there, the garrison).
- **Province taxes**: any of your provinces can have its own tax rate instead of the realm's; its unrest follows its own rate.
- **Remedies for an empty treasury**, offered only when it is in trouble (in debt, or short of a year's army and administration): debase the coin (a year's income now; customs -20% and +5% unrest for ten years), borrow the temple treasures (half a year's income; +15% unrest for five years), sell the tax farms (40% of a year's income; taxes -10% for five years; needs three provinces), a levy on the rich (30%; +10% unrest for a year), or ask a friend (an ally or overlord at +30 opinion may send silver). The silver pays off debt first.

Technology, administrative reach, corruption, rivals and score categories (built, 26 Sep 2026)

Decisions "Playable 21-26".

- **Technology web** (`data/techs.json`): 732 technologies in 15 branches, each running from 300 BC to today: farming, industry, metals and chemistry, energy, building and cities, transport, trade and finance, writing and communication, land and air warfare, seafaring and navies, government and law, society and ideas, science, medicine, and religion and the arts. Nothing invented before 300 BC is a technology: it is already part of the world at the start. Each has a year it becomes known to the world, prerequisites, a cost that grows with its year, and effects (crop yields, crafts, trade, tax, corruption, interest, manpower, garrisons, sieges, reach, sea travel, health, integration, unrest, admin capacity). Units and buildings can require a tech.
- **Research**: research points a year = (6 + town people/15,000 + organized people/300,000) x (1 + research bonuses). The player picks a tech in the Research panel (T key); bots choose their own.
- **Administrative reach**: provinces more than 700 km from the capital pay less tax and are less loyal. Admin capacity (from techs) sets how many provinces a realm runs well.
- **Corruption**: 5% + 25% x overreach (provinces beyond capacity) minus tech; it eats that share of tax income.
- **Rivals**: a powerful faction grows with corruption and overreach; the player can win them over with silver ("Win them over" in Treasury).
- **Score by category** (like Stellaris): People, Land, Wealth, Culture, Dynasty and Goals, shown in the Goals panel with a ranking of realms.

Plagues, disasters and each realm's coin (built, 26 Sep 2026)

Decisions "Playable 4" and "28".

- **History's disasters** (`data/disasters.json`): 53 plagues, earthquakes, eruptions and famines strike at their real dates and places: the Rhodes earthquake (226 BC), Vesuvius (AD 79), the Antonine Plague (165), the Plague of Justinian (541), the Black Death (1347), Lisbon (1755), the Spanish flu (1918) and more. Each has a centre, a reach in km and the share it kills at the centre, fading toward the edge.
- **Chance**: every region may suffer an earthquake each year if it lies on a quake belt (the Aegean, Anatolia, the Levant, Italy, Iran…), and anywhere may see an epidemic, more likely where many people live.
- **Medicine** cuts plague deaths (up to 90%), from the owner's health technologies.
- The dead are taken from the map's people; the land fills again over the following decades. A disaster in your lands stops a long turn.
- **Coins** (`data/coins.json`): money is silver by weight, and no talents appear anywhere in the game. Every sum shows in your realm's own coin: Roman didrachms, then denarii from 211 BC (lighter over time), antoniniani, gold solidi; Punic shekels; Ptolemaic (14.2 g) and Attic-weight (17.2 g) tetradrachms; Nabataean drachms; Gallic gold staters; Kushite gold by weight. The Treasury panel shows the silver in kg, what one of your coins holds in grams, and what it is worth in your trading partners' coins by the silver in each: if a didrachm holds 7.3 g and a tetradrachm 17.2 g, 1 didrachm = 0.42 tetradrachms. Gold coins count at 12 times their weight in silver.

Trade routes (built, 26 Sep 2026)

Decision "Playable 8".

- **Routes** (`data/trade_routes.json`): 35 historical routes, each a chain of real ports and towns: sea lanes (the western lane, the Tyrrhenian lane, the Sicilian channel, the Aegean, the Levant, the Straits and the Black Sea, the Adriatic, the Red Sea, the Gulf, the Atlantic coast), roads (the Incense Road, the Royal Road, the Great Road to the east, the Amber Road, the Rhone tin road, the Via Appia), and rivers (the Nile, the Tigris and Euphrates, the Danube). Later routes open in their year: the Via Egnatia (146 BC), the Venetian galleys (1000), the Genoese Black Sea route (1261), the Suez Canal (1869).
- **Who trades with whom**: a realm holds a place on a route if its land lies within 45 km of it. Realms linked by a shared border or a shared route trade directly; goods also pass along chains of up to three links through middlemen. Realms at war don't trade.
- **Tolls**: each realm that goods pass through (a middleman, or the holder of the places between buyer and seller on a route) takes 2% of their value. Holding a busy crossroads pays.
- **On the map**: the "Trade routes" map view draws them: sea lanes pale blue, roads gold, rivers blue-green. The Goods panel lists your routes and your tolls.

Town names from Pleiades (built, 26 Sep 2026)

Decision "Playable 23".

- `data/places.json`: 6,203 ancient settlements from the Pleiades gazetteer (CC BY 3.0, credited on the main menu), each with its names over time. In any year a town shows the name used then: Poseidonia in 300 BC, Paestum under Rome; Maleventum before it became Beneventum. About 3,600 stand in 300 BC.
- **On the map**: zoom in past the province names and the towns appear, the most important first (cities, places linked to many others, and places where many people live), never overlapping.
- The province panel lists a province's chief towns, and chance disasters are named after the nearest town.

Our own 300 BC borders, 56 realms, any realm playable (built, 26 Sep 2026)

Decisions "Playable 1", "20" and "30".

- **Borders**: `data/realms_bc300.json` lists 56 realms and the 613 historical districts they held in 300 BC. `tools/build_realm_mask.py` grows every district over the land at once, slowed by mountains, marsh and desert and barely crossing narrow straits, until it meets a neighbour or runs out of reach. Land no one reaches (the Sahara, inner Arabia, the high Alps) stays unclaimed. The old third-party (GPL) border data is gone. The race runs on a coarse grid, then each realm's share is blurred slightly and scaled up smoothly, so borders are curves, not stairs.
- **Capitals**: every realm's seat has a gold star on the map, and its name ("★ Roma", "★ Pella", "★ Meroe") appears as soon as you zoom in a little, before other towns, beside the star wherever there is room. A realm that loses its historical capital moves its seat to its most populous place. The province panel lists the capital first among its towns.
- **Floating panels**: every panel (the chronicle, the province panel, the treasury, goods, diplomacy, research, armies, goals…) can be dragged by any empty spot to anywhere on the screen, comes to the front when clicked, and keeps its place between games.
- **The world after Ipsus (301 BC)**: Antigonos is dead; his son Demetrios keeps a sea kingdom (Cyprus, Tyre, Sidon, Corinth, the Peloponnese garrisons, the islands). Lysimachos holds Thrace and western Asia Minor; Seleukos Syria, Mesopotamia and Iran; Ptolemy Egypt, Cyrenaica and Coele-Syria; Kassandros Macedon, Thessaly and central Greece (with Cilicia under his brother).
- **The new realms**: in Italy the Samnites, the Etruscan League, the Cisalpine Gauls, the Veneti, the Ligurians, the Lucanians and Bruttians, the Messapians, Tarentum and Syracuse under Agathokles; in the west Massalia, the Celts of the Danube, and the Iberian, Celtiberian and Lusitanian peoples; in Africa the Massylian and Masaesylian Numidians, the Mauri and the Garamantes; in Greece and the Balkans the Aetolian League, Sparta, Epirus, Rhodes, the Cretan cities, the Illyrian kingdom, the Dardanians, Paeonia, the Odrysians, the Getae and Byzantium; in Asia Minor Bithynia, Heraclea Pontica, Pontus, Cappadocia and Armenia; Atropatene; Kartli, Colchis and Caucasian Albania; the Bosporan kingdom and the Sarmatians; in Arabia Lihyan, Saba, Ma'in, Qataban and Hadramawt.
- **Rulers**: real where the sources name them (Agathokles and Theoxene, Seuthes III and Berenike with their four sons, Zipoites, Ariarathes II, Parnavaz, Spartokos III, Dromichaites, Glaukias, Audoleon…). Where no one is named (most tribes and republics) the ruler stands for the leading family with a name of the culture.
- **Cultures**: each realm has its own people from `data/cultures.json` (Italic, Etruscan, Thracian, Illyrian, Berber, Cappadocian, Armenian, Median, Colchian, Sabaean and more), and 16 naming traditions give children and new houses fitting names.
- **Any realm playable**: the start screen lists all 56, grouped by region, scrolling. The 19 tribal peoples start without provinces.

- **Armies, treasuries, treaties** for the new realms are in `data/start_realms.json`; new history goals include the Third Samnite War, Pyrrhus in Italy, Lysimachos against the Getae, the Illyrian, Jugurthine and Mithridatic wars.
- A people with no unit of its own for a role hires one from a people who fight that way.

Playability, part 1 (built, 28 Sep 2026)

Section 15, decisions "Next 41-46".

- **Outliner** ("Needs you", right side, floating and collapsible with its ▾ button): offers waiting, wars and their score, sieges and their progress, armies, provinces near revolt, research left idle, debt. Click a line to go there.
- **Hover cards**: rest the mouse on a province for a moment to see its owner, people, culture and faith, loyalty, unrest, garrison and chief towns.
- **Six more map views** (the menu at the bottom right): Culture, Religion, Loyalty, Wealth (what a person makes in a year under their ruler), People (people per km²) and Diplomacy (you, allies, vassals, enemies, truces, rivals, others). Tribal lands show their region's culture and religion. Each view has a key at the top left.
- **Realms table** (L, or "Compare realms" in Goals): every realm's people, land, silver (in your coin), income, soldiers, ships, techs and score, sortable by any column, with a graph of you and the five leaders over the years (people, silver, soldiers or score, recorded every 5 years and kept in saves).
- **Learning to play**: the first time each panel opens, a short tip explains it (once). "Guided first game" in How to play opens ten steps (taxes, research, a building, recruiting, a map view, a year, pretexts, a war, a conquest plan, a province won or peace), each ticked off as you do it.
- **The history of your realm** (decision "Next 47"): when AD 2000 arrives or your realm falls, a closing screen shows your score by category, your rulers in turn, your height, and your people and score over the centuries (also any time from Goals: "Your history so far"). **Closeness to history** (the designer's note): whenever one of history's campaigns (`history_goals.json`) reaches its end date, the game checks whether it turned out as it did (the attacker holds most of the regions' people; for campaigns that failed in history, like Pyrrhus in Italy, that it doesn't); the score is the share that did.
- **Data docs** (decision "Next 51"): DATA.md explains every data file in plain words, and the Data test checks that they agree with each other.
- **Time passing** (decision "Next 43"): beside Advance Year, a play button (or Space) lets the months run (1, 2, 4 or 8 a second); the date shows the month; each January the year is played; time pauses by itself when something needs you (a war, a succession, a revolt, a disaster, the end). Advance Year still plays whole turns. Armies will march month by month with section 11.
- **The look changes with the ages** (decision "Next 49"): Antiquity (Cinzel, bronze and gold), Late Antiquity from AD 250 (Cinzel Decorative, imperial purple), the Middle Ages from 600 (blackletter headings, EB Garamond, red and gold), the Early Modern Age from 1450 (IM Fell English, ink blue and brass) and the Modern Age from 1800 (Josefin Sans and Source Serif, steel). All fonts are SIL Open Font License (licences in assets/fonts). The chronicle marks each new age.
- **Music**: the Score Book (https://claude.ai/artifact/2bQczf4k9RVw2oiv59RfS2) lists the 49 tracks the game needs, for the designer to compose.

The economy rebalanced (built, 28 Sep 2026)

A test game showed Rome clearing some 5 million didrachms a year and piling up over 100 million by 280 BC. Three things were wrong, and are fixed:

- **Soldiers cost what they really cost.** Unit upkeep now covers pay and rations together (about 1.6 times the old figures): a Roman legionary drew 2 obols a day, 120 denarii a year, with his grain on top (Polybius 6.39); a Hellenistic mercenary about a drachma a day. Warships keep their lower upkeep, as fleets lay in their ship-sheds in peace.
- **The court, the gods and the city.** A new yearly expense, "Court, temples and public works": a fifth of the tax taken (palace, festivals, envoys and gifts) plus 0.1 talent per 1,000 people (roads, walls, granaries, magistrates). Ancient states spent nearly everything they took.
- **Not every state could tax alike** ("tax reach" in start_realms.json). The Ptolemies' royal monopolies and land tax (x1.8; Egypt now takes in about 6,300 talents a year), the Seleucids and Carthage (x1.5) took heavily. Republican Rome's citizens paid only the tributum, a war levy, so Rome takes 0.4 of the usual tax and lived, as it did, on booty, public land and its allies' soldiers.

At the start Rome now clears about 145 talents a year (roughly a fifth of its income), Egypt about 1,300, and most realms run close to even at peace strength. The accounts panel shows the same figures the game uses, techs and corruption included.

Battles and generals, part 1 (built, 28 Sep 2026)

The first part of section 11 (decisions "Next 1-10"): generals, battles that weigh the ground and the arms, battle reports and veterans. Marching by months, supply, fog of war, the tactical battle map, blockades, mercenary companies and reforms follow.

- **Generals** (data/generals.json). Every army has one. Five skills, 1-10: tactics (each point above the enemy's general is worth about 7% on the day), leadership (fewer losses), siegecraft (6% faster sieges a point), logistics (less weariness and siege sickness) and seamanship. Perks add to them or to an arm or a ground: Master of horse, Hill fighter, Master of elephants, Cautious, Reckless, Engineer, and more; four are for great captains only (Genius of war, Father of his soldiers, Stormer of cities, Lord of the sea).
- **Who leads.** The men of the ruling family (18-65) first. When the family runs short the realm appoints commanders, and each year it may meet "a man of great prowess" and raise him (3%, doubled at war); about one in eight is one of the great captains of the age. The great commanders of 300 BC lead from the start: Pyrrhos, Demetrios the Besieger, Seleukos with his elephants, Lysimachos, Ptolemaios and Agathokles. Generals age, retire at 70, and may fall when beaten (8%). In the Armies panel, "Change general" picks another general, a family member, or appoints a new man (20 talents).
- **Battles.** Each arm's strength is changed by the ground where the armies meet (from the land layer: hills, woods, marsh, desert, rivers; cavalry and elephants suffer in the mountains, light troops thrive), by what the enemy brings (horses will not face elephants, as at Ipsus; javelins turn elephants, as at Zama), by the general, by weariness and by veterans. The attacker must cross any river. The beaten side loses about 15% (more to a strong cavalry pursuit), the winner about 5%.
- **Battle reports.** Every battle is kept (the last 300, saved): place, ground, generals, men and losses, the odds and the day in phases. The panel opens by itself after a battle of yours; "Battle reports" in the Armies panel opens it any time.
- **Veterans.** Winning a battle hardens an army (+15% experience, more with some generals); a fully veteran army fights 30% better. Raw recruits dilute it.

Battles and generals, part 2: marching, supply and fog (built, 28 Sep 2026)

- **Marching by months** (decision "Next 3"). Armies no longer jump: each has a route and moves along it every month, about 90 km a month on open ground in summer (20 km a day on the marching days, with halts to forage), half that in winter (November to February). Mountains, marsh, forest, desert and river crossings cost more; weariness slows, a good quartermaster (logistics) speeds. Routes go through your land, your allies' and vassals', your enemies' at war, and land no realm holds; the land of other realms is closed. An army with warships crosses the sea.
- **Plotting a route.** In the Armies panel press Move, then click where the army should go; hold Shift to add stops on the way. The route is drawn on the map; the panel shows where it is going and about how many months it will take; Halt stops it. Painting a conquest sends the army marching to the place, and a siege only advances in the months the army stands before it.
- **Time runs month by month.** With the clock playing (Space), armies move each month; the turn button marches the rest of the year at once.
- **Supply and baggage** (decision "Next 4"). In your own or friendly land an army is fed and its baggage fills (up to 4 months of food). Within about 120 km of your borders supply lines feed it (further with a good quartermaster). Beyond, it forages from the people around it (little in empty land, less in winter) and eats its baggage; when the baggage is gone it starves, losing 4% a month and growing weary.
- **Armies meet.** Armies of realms at war within about 25 km of each other fight a battle (the one on the move attacks). The beaten army gives up its sieges and falls back toward its capital.
- **What your realm knows** (decision "Next 7", partial fog). Your realm sees its own, its allies' and vassals' land, 150 km beyond their borders, and 120 km around its armies. The rest of the land is dimmed, and enemy armies there are not shown.

Battles and generals, part 3: the battle map, sea battles, blockades (built, 28 Sep 2026)

- **The battle map** (decision "Next 1"). When your army meets an enemy, time stops and the battle map opens: a 20 by 12 grid whose hills, woods, marsh and river come from the ground where the armies met. Each side's troops stand in blocks (heavy foot in the centre, light troops and archers before it, horse and elephants on the wings), with a strength bar (gold) and a morale bar (blue).
- **Playing it.** Click one of your blocks, then a lit square to move it, or an enemy ringed in red to strike. Each block may move, then strike, once a turn: foot 2 squares, light troops 3, horse 4; archers and slingers shoot 3 squares, horse archers 2. Blocks can't slip past an enemy beside them. End turn lets the enemy move. "Let the general decide" plays the rest for you.
- **What decides it.** The old matchups (horse rides down skirmishers and struggles against formed foot, elephants scatter horse, javelins turn elephants), the ground (hills shield those on them, forest hides light troops and hampers horse, marsh and rivers weaken), flanks (each other block already on the target adds 30%, horse into a flank 50% more), the generals' tactics, veterans and weariness. Losses shake morale; a block with little morale or strength left flees, shaking those beside it. A side breaks when less than 30% of it still stands; after 15 rounds the stronger side keeps the field, and a drawn day is the defender's.
- **After the battle** the losses on the map become the armies' losses (fewer under a good leader, more for the routed), and the day goes into the battle reports.
- **Sea battles** (decision "Next 5"). Fleets that meet fight at sea: an all-water battle map for yours, the admirals' seamanship for the rest. Fleets can be sent out to sea like armies.
- **Blockades.** Enemy warships at least half as strong as a realm's own fleet, standing off its coast, cut a quarter of its sea trade each (at most 80%); the year's customs fall by the average. A besieging army with ships takes a coastal town half again as fast.
- **Setting.** Settings > Gameplay > "Fight battles yourself" (on). Off, your generals fight every battle.

Trade: your follow-up answers (built, 2 Oct 2026)

- **Markets** ("A market in each great city"): 20 of history's great markets (data/markets.json) open in their years: Rome, Alexandria, Carthage (until 146 BC, again from 29 BC), Syracuse, Rhodes, Delos (the free port, 166-88 BC), Antioch, Seleucia on the Tigris, Corinth (until 146 BC, again from 44 BC), Athens and the Piraeus, Ephesus, Byzantion, Tyre, Massalia, Gades, Puteoli (from 194 BC), Aquileia (from 181 BC), Petra. Each place belongs to the nearest open market; each realm's people are shared among its markets. Each market sets its own price for each good from what the realms it serves spare and lack; a realm buys and sells at its markets' prices (a trade between two realms at the average of both). In 300 BC 17 goods are priced differently by market (lentils from 0.5 to 3 times their usual price).
- **Merchants** ("You send merchants to markets to steer trade to you"): one merchant, plus one for each open great market in your land (at most five). Each takes 2% of its market's trade as customs, shared with the other realms' merchants there. Other realms send theirs to the busiest markets where their people trade. (Goods panel > Markets.)
- **Price shocks** ("shown in the chronicle"): grain prices at a market of yours rising to 1.4 times last year's are told, with the likely cause (enemy fleets, war, or poor harvests).
- **Price limits** ("Price limits by law, with shortages and black markets"): until laws exist, an edict in the Goods panel. Grain is bought at no more than its usual price; sellers sell less (the realm goes short by the price's excess) and the shortage adds 5% unrest.
- **Embargoes and trade treaties** ("as diplomatic actions"; Diplomacy panel): an embargo stops all trade between the two realms, even through middlemen; a treaty lets two realms trade directly with no border or route. They accept a treaty unless their opinion of you is below -20.
- **Pirates** ("Pirate havens that raid shipping and coasts until suppressed"; data/pirates.json): the Illyrians (until 168 BC), the Cretans (until 67 BC), the Cilicians (140-67 BC), the Barbary corsairs (1500-1830). A realm's customs fall by 30% of the share of its sea-route places within a haven's reach; coasts within reach lose 0.2% of their people a year. A haven is suppressed when its land is conquered or warships patrol within 200 km of it for six months.
- **Asked on the Ledger**: how your actions bend route shifts (your note), the number of merchants and what they do, other realms' treaties and embargoes, and how pirates are suppressed.
- **Later**: trade shown as moving lines thick by value, and visible caravans and fleets (the map's drawing); chartered companies (later ages).

Treasury and money: your follow-up answers (built, 1 Oct 2026)

Treasury > "Spending, coin and loans" opens the Finances panel.

- **Spending lines** ("You set each line"): court, temples and festivals, games, roads and works, grain; each 0% to 200% in steps of 25%. 100% is what a realm usually spends (court 10% of tax, temples 6%, games 4%, works 0.1 talent per 1,000 people, grain 4% of tax), so the old balance is kept; grain starts at 0%.
- Court: each step below 100% lets the rival's faction grow (4 points a year at 0%), above shrinks it.
- Temples and festivals: -5% unrest everywhere at 200%, +5% at 0%.
- Games: -3% / +3% unrest.
- Roads and works: growth (the burden factor +0.15 at 200%).
- Grain: -3% unrest and +0.1 growth per 100%.
- **Tax collectors** ("You choose tax farmers or salaried officials"). Officials: the usual administration cost; corruption takes its share. Farmers: pay 30% of a year's tax up front for a 5-year contract (it can't be broken sooner), keep 15% of what they collect, administration costs 60% of usual, corruption doesn't touch your taxes; +5% unrest and -0.2 growth.
- **The coin** ("You set the coin's silver content each year"): 30% to 100% silver. Prices move 15% of the way toward 1 / silver each year; all sums in your coin follow prices. Mint profit each year = tax x (1 / silver
- prices), large at first and fading as prices catch up. Customs fall with the silver ("Other realms value your coin by its real silver"). Rising prices add unrest (+10% per doubling). Restoring the coin costs (new - old) / new of a year's tax, at once.
- **Lenders** ("Several lenders, each with its own rate, limit and anger at default"): a shortfall is borrowed from the temples (6%, up to a year's income), then the bankers (12%, up to two years'). Surpluses repay the dearest first. Two years unable to pay or borrow and the realm **defaults** ("each lender reacts in its own way"): the temples' curse (+10% unrest and -0.3 growth for 10 years, your note: more than symbolic); the bankers refuse it for 30 years; realms it owed get a grievance (the "Unpaid debts" pretext for 20 years).
- **Lending to others** (Diplomacy > Lend): up to half your treasury, at most two years of their income, at 10%; they take it only when short of silver; they pay the interest and a tenth of the sum each year.
- **War indemnities** ("Yearly payments over many years"): a peace with tribute now sets an indemnity of two years of the loser's income, paid over ten years. Missing two years' payments, or you pressing "Stop paying", gives the winner the "Unpaid debts" pretext.
- **Money through the ages**: compared by silver until 1870, by gold until 1971, then by a basket of goods.
- **Retired remedies**: "Debase the coin" and "Sell the tax farms" left the emergency list; they are the coin and the collectors above.
- **Waiting for other topics**: who carries the tax (law and classes); the family purse (playing a family); officials stealing (events); banks, bonds and national debt (later ages); the full budget by province

with graphs (interface).

- **Not built yet, nothing in the way**: state monopolies on goods; each realm's coin changing as history changed it; the long list of history's remedies, each only in the years it was used (your note). These need research into the sources first.

Units of their own, the roster, and War: manpower (built, 1 Oct 2026)

Your answers: units of any size (you set 100 to 5,000 men), merging and splitting; armies hold an even number of units and splits make even numbers; experience averaged by men when units merge; a roster with each unit's strength, experience and origin; units raised where their people are; the dead are lost to the population, taken from the provinces the units were raised in; the wounded separate from the dead.

- **A unit** has a type, 100 to 5,000 men, its own experience (0 raw to 1 hardened), the province it was raised in, and its wounded. A new unit is its type's usual size (1,000 foot, 500 horse…).
- **Splitting** halves a unit (both halves at least 100 men). **Merging** joins two units of one kind (at most 5,000 men), experience averaged by men.
- **An even number of units.** After any change (recruiting, losses, disbanding, your own split or merge) the game splits the army's largest unit in two, or merges its two smallest of one kind, so the count stays even. A split or merge by you therefore changes the count by two. No men are lost or gained by this.
- **Losses** take a share of each unit's men. Of a beaten army's losses half die (the slaughter of the rout), of a victor's a quarter; the rest are wounded. Men lost with sunken ships, to hunger or to the sickness of siege camps mostly die (80%, 80%, 60%).
- **The wounded** heal: each month a quarter of them return to their unit and 3% die of their wounds.
- **The dead** leave their home province's people at the end of the month (spread over its towns and villages by their size); dead with no recorded home come from the capital province.
- **Where units are raised**: the province the army stands in, if it is yours; otherwise your capital's province. (Marching recruits to their army and training time wait on "Recruits marching to the army" and "Training time".)
- **Remnants** under 100 men join another unit of their kind, or go home.
- **The roster** (Armies panel): one line per unit with its men, wounded, experience and home, and Split, Merge and Send home.
- **Mercenary companies** hold units too, which keep their men and experience from employer to employer.
- **Unpaid desertion** takes men from every unit (it used to take whole units).
- **Who serves** ("a share per culture from history") waits on social classes; the levy share by free, dependent and enslaved people stays.
- **Speed tripwire** (your choice): the test of other realms fails if a year's calculations pass 1 second. It was 600 ms, a number I chose myself. When it trips, the code is made faster or you raise it; it is never a reason to cut simulation.
- **Speed**: supply distances are now worked out once a year per realm (borders change only when sieges end or peace is made), which brought a year from about 600 to 435 ms in the test of other realms.

Technology and knowledge: your follow-up answers (built, 30 Sep 2026)

Your answers on the Ledger's "Technology and knowledge" topic.

- **Research by branch** ("Researching several at once": a share of effort for each branch). Each of the 15 branches studies its own technology. You set each branch's effort with a slider (0 to 10) in the Research panel; the year's points are shared out in proportion. A branch at 0 gathers nothing. Other realms put all their effort into one branch at a time, the one with the cheapest open technology.
- **How far ahead of history** (open about 50 years early, dearer the earlier). A technology opens 50 years before the year the world knew it. At the earliest it costs 3 times its price, falling evenly to its price in its own year.
- **Learning from neighbours** (cheaper when neighbours and trade partners know it). A realm's contacts are the realms it borders and those on a trade route with it, not at war. A technology every contact knows costs 40% less (a share of them, a share of that).
- **Where research comes from** (towns, libraries, schools, great people, trade contacts). Each contact adds 4% research, up to 40%. Your note (education is a large part; people listed by education) waits on social classes (Population and people).
- **Inventions from beyond the map** (arrive on history's dates, spread along trade, researched after arrival). Paper (AD 750, by Samarkand), silk-weaving (AD 550), the stirrup (AD 600, with the Avars), gunpowder (AD 1250, with the Mongols) and the magnetic compass (AD 1190, by the Indian Ocean) can be studied only after their date, and only by a realm holding a region they enter by (Hyrcania, Media, Persis, Scythia, Pannonia, Arabia Felix) or bordering or trading with a realm that knows them.
- **Who knows a tech** (each region knows it; conquest brings what they knew). Taking a province teaches the conqueror some of what its old master knew: each technology the loser knew whose prerequisites the conqueror has, with a chance of twice the share of the loser's people taken (5% to 60%). The chronicle names what was learnt.
- **Pace** (near history's pace; leaders a few decades ahead). Old knowledge gets cheaper as it spreads: a technology costs up to 50% less, reached 200 years after its date. All research is scaled by 1.5, set by a 100-year run from 300 BC: by 200 BC Egypt (the leader) knew 77 of the 78 technologies the world knew and had one ahead of its time; Carthage 53, the Seleucids 39, the Gauls 23.
- **Waiting for other systems**: knowledge lost when cities fall and literacy drops, and found again only as derivatives unless history truly recovered them (your note) — needs literacy (Population and people); great movements spreading like religions — needs Culture and religion; spies stealing techs — needs Diplomacy; the tree's look ("Both") — a tree and a timeline view, with the interface work.

The Bronze Age and the opening

(DECIDED IN PART, 11 OCT 2026)

Your wish (11 Oct 2026): to play in the Bronze Age, above all the collapse, Ugarit, the Hittites and the Hattians. Your answer on how the game begins:

> The beginning should be a simulation/history lesson that spans the earliest gatherings of people for one purpose such as Göbekli Tepe, and then go through the early settlements like Çatalhöyük, then starting the game at the point settlements became polities. It should be like a cutscene.

Questions waiting on the Ledger (topic "The Bronze Age"): How the Bronze Age fits in; When settlements become polities; What the opening shows; Watching the opening; Is the opening simulated?; How the collapse plays; What makes the Bronze Age different; When to build the Bronze Age.

Buildings and works, part three: roads (built, 11 Oct 2026)

From your answers on the Ledger (How roads are drawn, Kinds of road, What roads do, Roads others build, Which kinds of road, What a road costs, Enemies on your roads):

- **Kinds by material** (`data/roads.json`), each faster and carrying more: earth road, gravel road, paved stone road (needs Surveying with the groma), macadam road (Macadam roads, 1820) and asphalt road (Motorways, 1930). Railways are not roads and come with their own network later.
- **Drawing one**: in a province's Buildings tab, pick a kind under "Draw a road", click one of your towns, then another. The road finds the easiest ground through your own land. A bar shows its length, bridges, silver, workers, years and upkeep, with Build and Cancel; right-click stops drawing.
- **Cost**: silver per km (more over rough ground) plus each bridge; workers per km taken from manpower while it is built; years at 100 km a year, up to four times faster as the realm learns transport and construction (your note on finding a formula for building faster through the ages); upkeep each year of 1% of its price.
- **What roads do**: everyone marches faster on them, invaders too (a paved road halves the ground's cost whatever the ground); the provinces they cross trade more; a road all the way to the capital takes much of the far-province penalty away; ideas travel along them (up to +10% research at 5,000 km of road). Disease travelling along roads waits on "Epidemics that travel" (Population and people).
- **Breaking your own**: each of your roads has a Break button (it speeds no one) and Repair (a quarter of its price, a year).
- **Neglect**: three years of unpaid upkeep and a road falls a kind (paved to gravel to earth to nothing).
- **History's roads** (`data/roads_history.json`, 26): those older than 300 BC stand at the start (the Royal Road, the Via Latina, Via Appia and Via Valeria); later ones are laid by their builders on their dates, a leg at a time where they hold both ends (Rome's viae, the Via Egnatia, Trajan's roads). Railways among them wait for the railway network.

Buildings and works, part three: canals, great works and frontier walls (built, 11 Oct 2026)

From your answers (Canals and great works, Which canals and great works, Great works others build, Frontier walls, How frontier walls are drawn, What frontier walls do, History's frontier walls):

- **History's canals and great works** (`data/works_history.json`, 15) are built as wonders are: each once in the world, history's builder (or whoever holds the place) begins it on its date, you can race them from that date, hurry them, and they fall on history's dates (the Canal of the Pharaohs silting up). They give realm effects: the canals trade and cheaper sailing (Suez +15% trade, -10% sea travel), the drains and dams more field crops (the Fucine Lake +3%, the Aswan High Dam +8%). The Diolkos of Corinth stands in 300 BC. They are listed with the wonders in a province's Buildings tab.
- **Canals you dig** (`data/roads.json`, "canals"): pick a kind under "Dig a canal", click your land by the sea or a river, then by another water. Without the pound lock (1373) a canal can't climb hills; with it, a canal with locks can; Great ship canals (1869) open the ship canal. A canal costs silver, workers and years like a road, brings trade to the provinces it crosses, and your ships sail through it. Unkept, it silts up.
- **Frontier walls you draw** (`data/roads.json`, "walls"): ditch and rampart, or stone wall. Pick one under "Draw a frontier wall", click your land where it starts and keep clicking along your border; each click adds a stretch, then Build. Once finished it takes its garrison from manpower (25 or 40 men a km); unmanned it decays in three years. Armies of another realm (not an ally) lose 40 or 90 km of march for each crossing; provinces along it are better held (garrison +1 or +2). Stopping raids waits on raiding (the campaign question "Raids").
- **History's walls**: the Long Walls of Athens stand in 300 BC and fall to Sulla in 86 BC; Rome builds the Fossatum Africae, the Strata Diocletiana, the Hexamilion and the Anastasian Wall on their dates where it holds both ends. Walls whose builders aren't realms of the game (Gorgan, Derbent) aren't built by others (your answer: only their own history's).

Your answers of 11 October 2026 (recorded)

92 answers from the Ledger. Built so far: the three wonder answers and the pirates' (below); the rest wait on their systems (see ROADMAP.md).

Buildings and works

- **Which canals and great works**: History's own plus any canal you draw between two waters.
- **Great works others build**: Yes, as wonders: only their own history's, and you can race them.
- **What a road costs**: Also workers taken from manpower, like wonders. Your note: "need to find formula for lessening time it takes as eras/years progress"
- **Enemies on your roads**: Everyone, but you can break your own roads to slow an invader.
- **Which kinds of road**: your own answer. Your note: "railway is not a road, and they should primarily be split by material and hence speed and bandwidth"
- **How frontier walls are drawn**: Click point to point along your border; cost by length and ground.
- **What frontier walls do**: Stop raids, slow armies crossing; need a garrison from manpower or they decay.
- **History's frontier walls**: Yes, only their own history's.
- **Hurrying a wonder**: More silver and workers make it faster. Your note: "Up to a quarter faster at triple the cost"
- **Wonders whose builders aren't in the game**: Whoever holds the place builds it on history's date.
- **Saving a wonder from its fall**: Ask me each time.

Cities and settlements

- **Moving the capital, in detail**: Also: administrative reach centres on the new capital at once.
- **What holds a city back, in detail**: Each city has a water limit (aqueducts raise it), a food limit (what its land and trade bring in) and disease that grows with size (baths and sewers lower it).
- **Choosing spare, plunder or raze**: Spare: loyalty; plunder: silver and slaves, buildings damaged; raze: the city becomes a ruin and others fear you.
- **City tiers**: your own answer. Your note: "At least 15 tiers"

Diplomacy

- **Client kings**: Clients keep their ruler and laws, pay tribute or troops; you can install or remove kings.
- **Demands and insults**: Send demands and insults.
- **How envoys work**: Envoys are characters who travel for weeks or months; their skill matters; you have a few.
- **Hostages**: Hostages guarantee treaties and learn your ways.
- **International bodies**: They appear as history had them and can bind members.

- **Leagues**: Members, a leader and votes on war; you can join, lead or break them.
- **Making peace**: Build a treaty from items the other side weighs.
- **Pretexts by era**: A full list by era.
- **Who you can deal with**: Only realms you know, widening with contacts and trade.
- **More kinds of treaty**: Non-aggression, military access, trade, subsidy, marriage alliance, hostage exchange, federation (foedus), defensive league.
- **Paying off invaders**: Pay tribute to buy peace from raiders.
- **Trust and fear**: Separate trust (do you keep your word) and fear (your might and aggression).
- **What can be given**: Silver, goods, famous objects, land, and help (troops, grain).
- **Protectorates**: You guarantee a realm's independence and may intervene in its affairs.

Government, laws and administration

- **Which estates**: By realm and era: Senate and equites, priests, army, cities, nobles, merchants, peasants.
- **The kinds of government**: Yes, once I've checked the draft.
- **How a law passes**: Each law costs some of the four points as well as time and silver.
- **What the points buy**: Authority: laws and reforms; Piety: religious acts; Loyalty: holding estates and armies; Legitimacy: succession and claims. Your note: "some are multifaceted"

Land, resources and goods

- **Adopting new crops**: A crop arrives by trade on history's date; you pay to spread it province by province. Your note: "Non-player realms adopt them whenever they historically do, if they never do historically then they don't in the game."
- **History's named mines**: About 30 of the great ones.
- **Prospecting**: Pay to survey a province once the technology is known; history's finds come on their dates for everyone.
- **How stores work**: Also sold when prices are high, by your order.

Population and people

- **Which social classes**: Nobles, priests, citizens, freedmen, peasants, slaves, foreigners, changing by era and culture.
- **What each class wants**: Each class needs a basket of goods; going short angers it.
- **How literacy rises**: Schools, libraries, towns, printing, religion; it falls in war and collapse.
- **Towns and countryside**: Towns from history's sizes (Rome about 300,000 by 1 BC), the rest countryside; people move to towns that have work.

Provinces

- **Province focuses**: Farming, herding, crafts, trade, mining, soldiers.
- **Choosing governors**: Characters you appoint; their skills change taxes, unrest and corruption; they may grow too strong.
- **The province kinds**: Core, conquered, allied city, client kingdom, temple-state, colony, frontier march.
- **Redrawing your provinces**: Split or join provinces in the province panel, at a cost of loyalty for a while.

Sound and music

- **Ambient sound**: Sounds of what's under the camera: sea, market, wind on the steppe.
- **Sound effects**: Clicks, coins, marching, battle, sea, crowds, varied by era.
- **How music is chosen**: By situation, culture and era, crossfading.
- **Separate volumes**: Music, effects and ambient apart.

The court: characters and families

- **What councillors do**: You give each a task (collect taxes, watch the court, win a people over) and their skill decides it.
- **Councillors' own will**: They have opinions, ambitions and may resign or plot.
- **The court screen**: A hall with the ruler, family, council and courtiers as coins.
- **Personal unions**: The junior realm keeps its laws and may break away when no common heir remains.
- **Marriages and treaties**: A royal marriage can seal an alliance, but doesn't make one by itself. Your note: "can be chosen as a law to automatically make an alliance"
- **Marrying across faiths and peoples**: By each culture's and era's rules.
- **Marriage offers**: Other courts send offers you can accept or refuse.
- **What a family controls**: Only what its members hold: offices, armies they command, their estates. Your note: "if a member holds a title to rule, the family is named as the ruling family and all close members holds the "royal" trait instead of their "noble" trait"
- **Family life when playing the realm**: The ruling family's events still reach you.
- **Switching mid-game**: You can switch between realm and family. Your note: "Let's find the best way to do this"
- **Finding plots**: The spymaster and loyal friends uncover plots; you learn the plotters.
- **Plots abroad**: Plot against foreign rulers and courts.
- **Punishing plotters**: Pardon, exile, imprison, execute, each with consequences. Your note: "at least 2 more options such as enslave or send to colosseum"
- **Who gets a portrait**: Everyone at court.

- **Ambitious regents**: A regent may try to keep power or take the throne.
- **Which magistrates**: Each republic's own offices and ladder of office.
- **Governors in republics**: Former magistrates govern provinces.
- **The Senate and the assemblies**: A senate and assemblies that vote on war, laws and commands.
- **Skills through life**: Skills grow with age, education and office, fade in old age.
- **How skills are shown**: Both. Your note: "use creative words and 1-100"
- **Co-rulers**: Raise an heir to co-ruler.
- **Disputed successions**: Rival claimants can start a war of succession.
- **Succession laws**: History's laws by culture: eldest son, any son chosen, brother first, elective, tanistry, shared kingdom.
- **Traits changing**: Traits come and go with life's events.
- **The family tree**: Filters: living, heirs, claimants, marriages; click to open anyone.

The map

- **Which coasts change**: Also drainage and reclamation you build.
- **The whole globe: order**: India, then Central Asia and China, then Africa, then the Americas.
- **What the map editor edits**: Borders, provinces, towns, names, starting realms.
- **Passes, fords and straits, in detail**: Armies must use them to cross mountains, rivers and straits; holding one blocks it; winter closes passes.

Time and turns

- **Finer time**: Months are fine. Your note: "add weeks"
- **What stops time**: A list in Settings where you tick what stops time.
- **Years per turn with the clock**: Keep both: the clock and the turn button.

Trade

- **Other realms' trade treaties and embargoes**: Yes, history's treaties on their dates, and their own by their aims.
- **How many merchants**: Both.
- **What a merchant earns**: your own answer. Your note: "Silver, Steer goods & bring news"
- **Suppressing pirates**: Need enough warships. Your note: "Suppressed havens return later also"
- **How your actions alter route shifts**: Both.

War: armies, units and generals

- **Which laws decide standing troops**: A number per realm from history (start_realms.json).

War: mercenaries

- **Captains who rise, in detail**: To a lordship or a throne, if they hold land and their men are loyal.

War: reforms

- **What makes a reform available**: History's causes plus a technology.
- **Who is angered by reforms**: Your rival's strength rises.
- **Which army reforms first**: Every reform we can source to 1 AD.
- **Refusing a reform**: You can take it later from a reforms list.

Built from these answers (11 Oct 2026)

- **Hurrying a wonder**: a wonder can be built up to a quarter faster at up to three times the silver and workers (the province panel offers a hurried button).
- **Wonders whose builders aren't in the game**: whoever holds the place begins it on history's date (up to 50 years late), with the goods it needs.
- **Saving a wonder from its fall**: when history's date comes, an offer appears in Diplomacy; the wonder stands while you decide. Accepting pays half its price and it stands for good; declining, or leaving the offer for two years, lets it fall.
- **Suppressing pirates**: each haven has a naval strength (Illyrians and Cretans 4, Cilicians 8, Barbary corsairs 10) that grows 1% a year unchecked, up to three times. Patrol months count only when your warships in reach are at least as strong. A suppressed haven returns 40 years later unless the realm holding its coast keeps warships there at least as strong as the pirates first were.

Buildings and works, part one (built, 4 Oct 2026)

From the designer's answers on the Ledger (Buildings and works):

- **Many buildings** (`data/buildings.json`, 80): the first twelve, one building for nearly every good (olive groves, potteries, vineyards, dye works and so on; each raises its good in the province by a share per level), and warehouses (spoilage), libraries and schools (research), baths, theatres, a mint and an agora.
- **Levels**: each building has named levels (harbour, then great harbour); level n costs n times the base price and takes the same years.
- **Queue**: up to 6 works waiting per province. One is built at a time; a town of 50,000 builds two at once, of 200,000 three.
- **Decay**: when upkeep goes unpaid for 3 years in a row, one building in each province loses a level; a province of under 1,000 people lets its buildings fall too.
- **War damage**: a province taken by siege loses a level of its walls, and each other building has a 30% chance to lose one; the chronicle says so.

Buildings and works, part two: wonders (built, 4 Oct 2026)

From the designer's answers on the Ledger (Buildings and works: which great works count as wonders, wonders others build, wonders destroyed, building a wonder, what wonders do):

- **The list** (`data/wonders.json`, 38): the Seven Wonders and wonders through the ages, from the Pharos and the Library of Alexandria to the Pantheon, Hagia Sophia, the Dome of the Rock, Notre-Dame, St Peter's, Versailles and the Eiffel Tower. Each has its real place, history's dates, its builder, cost, workers, special goods and effect.
- **Once in the world.** The Great Pyramid, the Hanging Gardens, the Temple of Artemis, the Statue of Zeus and the Mausoleum already stand in 300 BC.
- **No one before history.** Nobody can begin a wonder before the year history began it.
- **History's builders.** The realm that built it in history begins it on that date if it holds the place and has the special goods (up to 50 years late), and takes history's years. Other realms build only their own history's wonders. A wonder whose builder isn't a realm of the game (the Dome of the Rock) is left for you.
- **You race them.** From history's date you can begin any wonder in any of your provinces (a coast where it needs one). Whoever finishes first has it; the others' work stops, their workers come home and half their silver comes back. You can build several.
- **The cost**: silver, workers taken from your manpower while it is built (5% never come back), and special goods your realm must make or import (marble, granite, papyrus, bronze…). Work halts under siege and stops if the province is lost.
- **Effects** go to whoever holds the province: the Pharos makes sea travel cheaper, the Library brings a quarter more research, the Colosseum calms the people, Versailles lets the court manage more provinces. Each adds its culture to the score.
- **Falls.** Wonders fall on history's dates (the Colossus in 226 BC, the Temple of Artemis to the Goths in AD 262). Only yours can be saved: when the date comes you are asked in Diplomacy whether to save it for half its price (changed 11 Oct 2026, your answer "Ask me each time").

Battles and generals, part 4a: mercenary companies (built, 28 Sep 2026)

Your answers: companies replace "hire at twice the price"; history's companies in their eras plus generated ones; hired at hiring grounds (be near or send an envoy); 500 to 5,000 men of mixed arms; each captain a character; monthly pay plus a bonus after victories, strong companies with a price range that can bankrupt you; unpaid they desert, change sides or revolt; they go to whoever pays best. Each unit keeps its own experience.

- **Hiring grounds** (data/mercenaries.json): Cape Taenarum, Crete, Rhodes, the Balearics, Thrace, Galatia, the Alpine Celts, Celtiberia, Numidia, Libya, Messana, Illyria, Aetolia, Thessaly, Olbia, Tanais, Arabia, Persis.
- **Companies.** 22 from history, each in its years (Xanthippus' Spartans 257-250 BC, the Gaesatae 236-222 BC, the Mamertines 289-264 BC…). Each year there is a 30% chance a company is raised at a ground with none waiting, from the peoples there; it waits 6 years and then disbands.
- **Hiring.** Armies > "Mercenary companies". Within about 40 map steps (360 km) of your land or one of your armies you pay the first month; from afar an envoy costs one more month. The company becomes an army at its ground, led by its captain.
- **Pay.** Each month: the troops' own yearly cost ÷ 12 × the company's factor (1.3-1.5 for ordinary companies; elite ones name a new price each year inside their range, e.g. Xanthippus 1.6-3.5). Each victory adds 1-2 months' pay owed. Mercenaries are not in the army's upkeep; the treasury panel shows their year's pay as its own line.
- **Unpaid.** Owed 2+ months: a 15% chance each month they desert. Owed 4+ months: a 35% chance they go over to your richest enemy, or revolt, plunder up to 6 months' pay and march away.
- **Outbidding.** Offer 1.25× their pay (with two months' pay in hand) and a company leaves its employer for you, keeping its experience.
- **Other realms** at war hire the largest company near them if they can afford a year of it.
- **Recruiting your own** now always takes manpower; without enough men you are told to hire a company instead.
- **Experience by unit type.** New recruits dilute only their own kind of unit; the veterans beside them keep their edge.
- Fleets' commanders are shown as admirals, companies' as captains.

Dynasties and characters (decided, built)

IN SHORT

Ruling families live, marry, have children and die. The crown passes by real inheritance rules, and a new house takes over when a line dies out or a throne is seized. Built and in the game.

Depth for now: enough for ruling families to live, marry, have children and succeed each other for the whole game, without traits, events or intrigue (those can be layered on later). Code: `src/Dynasties/` (`CharacterRegistry.cs` for succession and mortality, `CharacterRegistry.Life.cs` for the yearly simulation, `Names.cs`), the 300 BC families in `MapView.RealCivs` / `FrontierZones`. Tests: `DynastyTest`, `DynastyLifeTest`, `TimeTest`.

- **Starting families.** Each realm starts with its real (or, where the record is thin, plausible) 300 BC ruling family with real ages: Antigonos the One-Eyed at 82 with his son Demetrios and grandson Antigonos Gonatas, Ptolemaios I at 67 with Berenike and their children, Seleukos with Antiochos, and so on. Rome, a republic, is represented by its leading house (the Valerii). Rulers range from 38 to 82.
- **Cultures.** Every realm and character has one of eight naming traditions (Latin, Punic, Greek, Meroitic, Nabataean, Iberian, Celtic, Scythian). Children take the culture of the parent whose house they belong to; a first son is sometimes named for his paternal grandfather and a first daughter for her maternal grandmother.
- **Mortality.** A pre-modern elite life table: 8% die in the first year, about one in six before five, 0.8% a year from 15 to 39, then a rise from 1% at 40 doubling about every nine years (cap 35%). About 63% of 40-year-olds reach 60 and a third reach 70. Average reign comes out around 26 years.
- **The court.** Only each realm's court marries and has children: the ruler, spouse, children and grandchildren, siblings with their children, and the heir's household. Everyone else lives out their life but the family tree stops branching there, which keeps the world to a few hundred living people however long the game runs.
- **Marriage.** Unmarried court members (men 17-55, women 15-38) marry with a 35% chance a year; three in ten matches are with another realm's court (the closest in age within 15 years, from another house), the rest with a generated noble of the realm's culture.
- **Births.** A married woman of 16-42 has a child with a chance of 24% a year in her twenties, falling to 3% after 40; at most nine children. About four or five births per full marriage.
- **Succession.** Primogeniture with representation: the ruler's descendants in order, the eldest line first, so a dead eldest son's children come before his younger brother (under male preference, sons' lines before daughters' at every level). With no descendants, the search moves up: brothers and nephews, then uncles and cousins, up to four generations. A princess who married outside the great houses keeps her children in her house and line. **No personal unions:** someone already reigning elsewhere is passed over, so realms never merge by inheritance (a later decision could allow it).

- **New houses.** A realm gets a new ruling house when its family has died out, or when a strongman seizes the throne at a succession: 6% at every succession, plus 25% when the heir is a child under 16, plus 10% when the heir isn't the late ruler's own descendant. The founder (30-50, of the realm's culture) comes with a wife and children. Over two thousand years, houses last about 280 years on average.
- **The chronicle** shows successions and new houses in every realm, and births, marriages and deaths only in the player's own court.
- **Known limits.** Dead characters are kept for family history, so the save grows with time (about 7 MB of family records by AD 1700). The HUD shows the heir's name but not why they're heir (a family tree view is future work).

New concepts needed

IN SHORT

A shopping list of things later systems will need: named settlements, capitals, resources, roads, Might, map overlays and the conquest panel.

- `Settlement` — named populated place: id, name, position, population, tier, founded year, plus a status (`active` / `ruin`) for the hysteresis and ruin rules in Population & settlement engine. Tracked separately from provinces: a province can contain several, and unclaimed or unorganized land carries whichever clusters clear the percentile threshold. Named status survives conquest and loss of ownership.
- **Capitals** — each realm has one country capital, and each organized province has one province capital (its administrative seat). Both are a settlement reference plus a "first designated" record per realm, so the one-time resettlement can't repeat. They drive the growth bonuses in Capital growth bonuses.
- **Resource categories** — coal, ores (iron/copper/tin/gold/silver/lead), water, flora/fauna (timber/game/fish/wild plants), plus categories that tie into the existing civ flavor text rather than inventing a separate trade- goods system later: Kush's iron industry, Nabatea's incense routes. (OPEN) full list not finalized.
- **Infrastructure tier per cell** — none/dirt/gravel/paved/rail/electrified, hard-capped at gravel while unorganized.
- **Might** — see the dedicated section below. Supersedes the standalone "local militia" idea by generalizing it into the domain-split strength score used everywhere annexation resolves, for both realms and raw land.
- **Map overlays** (UI) — a control to switch what the map draws on top of the terrain: Political (today's ownership view, the default), **Regions** (the ancient geographic regions and their names, the only place those names appear; built as a toggle), and later the population heatmap and resource overlays. One overlay active at a time.
- **Floating confirmation panel** (UI) — live Might + resource-icon readout while painting; see Annexation UX, below.

The five simulation systems (realm/province level)

IN SHORT

The five big systems a realm runs on: population, resources, economy, technology and military.

- **Population** (per province, summed from the density field) — how many people there are comes from natural increase (food security, disease, urban penalty, security, land and living standards, burden, health, climate), and where they live from attraction and bounded migration; see Growth drivers. Carries integration/sentiment (freshly conquered = low, decays toward assimilated over years unless suppressed by unrest).
- **Resources** (per province, summed from the density field) — yield type from terrain; gates military unit types and economic output.
- **Economy** (realm-level, aggregated from provinces) — treasury, income (tax from population x output) minus expenses (upkeep, admin cost).
- **Technology** (realm-level) — military / administrative / agriculture tracks; administrative tech is what should eventually cap how much territory a realm can hold before unrest/corruption penalties bite.
- **Military** (realm-level armies drawn from province manpower + resources) — recruited units gated by resources and tech; a leader/general slot extends the existing `Character.traits`, not a new character concept.

Might: the unified strength score

IN SHORT

One strength score per realm or piece of land, split into Land and Naval (Air and Space later), compared whenever someone tries to take land. Unowned land defends with its local population.

The number actually compared whenever annexation resolves. Split into domains that unlock by era: **Land** and **Naval** (both available from the game's 300 BC start — ancient/medieval navies were real, triremes and quinqueremes included, and several starting civs are explicitly maritime powers: Carthage, Rome, the Greek world, Ptolemaic Egypt), **Aerial** (~1900 AD), and eventually **Space**. A domain that hasn't unlocked yet shows as 0/greyed out for every realm equally. The data model carries all four slots from day one (cheap now, expensive to retrofit later) even though two sit inert for a long time. (DECIDED)

(The ~1500 AD instinct isn't wrong, just describes a different thing — true ocean-crossing "Age of Sail" capability, as opposed to Naval Might existing at all. That's a plausible later refinement, e.g. a coastal/riverine vs. blue-water distinction, not something blocking Naval Might being active from turn one.)

Where a Might number comes from depends on what's being measured: - **A realm's own Might** (the attacker, or an organized province's defender) — derived from that realm's Military system (armies, tech, fortification), per domain. - **Unclaimed or unorganized land's Might** (defender only — raw land has no attacker side) — derived directly from local population density: the same per-cell density field already planned under Layers, read live, not a separately baked asset. "Heat map" here describes the *display* (a colored intensity overlay), not a second data source that could drift out of sync with the density field it's reading.

This resolves both open questions from before: - Defense is **not** seat-anchored — it's the whole province's aggregate Might, so beating it (in whichever domain(s) apply — Land and Naval, for a very long time) flips the entire province regardless of exactly which cells got painted, matching the earlier "conquered a state from one city" decision. - Resolution reads as **instant at confirm-time** — one decisive Might comparison when you commit, not a multi-tick siege bar. (Inferred from "confirm… if you can handle it" rather than stated outright — flagging in case a multi-year siege was actually intended.)

One refinement folded back into Layers: since Might for unclaimed land is explicitly meant to reflect *historical* population, the density field itself is sourced from **HYDE** (History Database of the Global Environment — PBL Netherlands Environmental Assessment Agency / Utrecht University), which gives gridded population estimates from 10,000 BCE to the present (HYDE 3.2.1, CC BY 3.0). Free, public, no licensing cost — same footing as the Natural Earth coastline and historical-basemaps borders already in the project. Implementation follows the same pattern as `tools/build_land_mask.py`: a new `tools/build_population_mask.py`-style import step bakes HYDE's grid into textures on this project's own projection. One snapshot is baked per HYDE keyframe the game's timeline covers, starting at 300

BC, since the snapshots double as the `H_hist` keyframes in Population & settlement engine. Terrain-driven estimation is dropped as the source of record; it remains only as a plausible gap-filler if HYDE's resolution turns out too coarse for a specific region. **(DECIDED)**

Annexation UX: fog of war and the floating confirmation panel

IN SHORT

You can't see a target's strength until you start painting over it. Then a floating panel shows the strength and resources of exactly what you've painted, province by province, before you confirm.

A target's Might is hidden until you start painting over it — no defense numbers just from looking at the map. Starting a paint stroke on unclaimed or enemy territory triggers the reveal, and it updates live as the stroke grows or shrinks, reusing the same dirty-cell tracking the ownership proposal paint already has.

While painting, a floating confirmation panel shows, live: - Might per domain for whatever's currently painted (Land and Naval as real numbers; Aerial/Space greyed out at 0 until their era unlocks) - Icons for whichever resource types are present in the currently-painted cells

Confirm stays available regardless of whether you're favored — showing the numbers is about an informed choice, not a blocked action. Losing (per the existing rule below) means the paint snaps back with attrition to the attacker's own Might, not that confirm gets disabled. (DECIDED)

When one stroke spans multiple distinct provinces (or crosses from unclaimed land into an organized enemy province), the panel shows a **breakdown per province** — a separate Might/resource readout for each one touched, not a blended total. Matches how resolution itself happens: each affected province is checked against its own Might independently, so the panel should show exactly what's about to be decided, not an average that hides which individual provinces you can and can't take. (DECIDED)

Annexation resolution

IN SHORT

Conquest stays a brush stroke. On confirm, each province you painted is compared against your Might: win and you take the whole province, lose and the paint snaps back with losses to your army.

Stays a paint action, not a menu. But the *unit of capture is the province*, not the cell, once land is organized: painting/confirming inside an enemy province checks that province's Might (see above — garrison/fortification, derived from the owning realm's military + tech + that province's integration) against the attacker's Might. Beating it flips the *entire* province — every member cell, including ones the brush stroke never touched, keeps its existing `province_id`, population, infrastructure, and settlements, just under a new owning realm (mechanically: the province's owner pointer flips, then a bulk write propagates that to its cells, per the invariant above). Same brush, same immediate feedback — the simulation decides whether paint sticks, at province granularity, not a separate siege UI. (DECIDED)

Unorganized land (enemy-held, but never administered into a province) has no "whole shape" to inherit, so it still resolves locally — per painted cell or contiguous painted blob — against its population-scaled militia. Unclaimed land (no owner at all) resolves the same way, against its own native militia — usually weak given how sparse unclaimed populations tend to be, but not an unconditional freebie either.

This also settles the earlier open question about post-conquest organization: captured organized provinces do **not** reset to unorganized and don't need a separate "decide how to organize this" step. They arrive already organized, under new management, exactly as they were run before.

Since a province is now captured as an atomic whole, **province size is a real balancing lever**: small/numerous provinces make conquest gradual, a few huge provinces make it all-or-nothing and swingy. Worth weighing during Phase 1 (province generation/seeding), not just an implementation detail.

Suggested build order

IN SHORT

The order the systems are built in, each checked in the real game before the next begins. Step 1 (provinces) is done; step 2 (population) is partly done.

1. Province layer (`province_id` grid + brush-editable boundaries), no stats yet — verify rendering/paint behavior only.
2. Density fields (population + resource-type per cell), provinces sum on demand. Population via the engine: HYDE-sourced `H_hist`, `H_sim` seeded from it, blend, then `C * H^k`. Start with `H_sim` static and verify the 300 BC output against HYDE before adding any dynamics. Then the historical ceiling, with a player/non-player split, and capital bonuses; rerun `tools/calibrate_population.py` on the real grid. Then the Growth drivers groundwork: regional totals, conserving migration, and the balance-test harness, before any later system writes a factor score.
3. Unorganized-territory capability gating (the matrix above) + settlements (clustering, percentile naming, hysteresis, ruins), plus the regional floor. (The region mask and Regions overlay were built with the growth drivers, which needed them.)
4. Economy (realm treasury from province sums).
5. Military + tech (army/power score, minimal tech multipliers).
6. Integration/sentiment (decay-toward-assimilated stat).
7. Paint-gated annexation (the resolution function tying 2-6 together).

Status: population engine and growth-driver groundwork built. Built so far: `src/World/PopulationEngine.cs` (stages 1-2, the historical ceiling with player legacy fade, capital bonuses and resettlement, HYDE-based seeding and 25-year natural resettlement, save/load), the growth drivers' two channels with every factor at its baseline (`PopulationEngine.Growth.cs`), the 38 geographic regions (`tools/build_region_mask.py`, `PopulationEngine.Regions.cs`) and the Regions overlay (`MapView.Regions.cs`), `tools/build_population_mask.py` (the HYDE import), the engine wired into world generation, the yearly tick, saves, and a HUD population line, and each real civ's 300 BC capital registered as an existing capital. Covered by `src/Tests/PopulationEngineTest.cs` (toy-world mechanics), `src/Tests/PopulationHydeTest.cs` (the calibration on the real grid) and `src/Tests/PopulationBalanceTest.cs` (the growth drivers' balance tests). Not built yet: stage 3 (clusters, percentile naming, ruins' labels, the regional floor), gameplay systems that write factor scores and attraction (the engine exposes `SetFactor` and `DriverMods` for them; until they exist the world follows history), events that trigger `ApplyShock`, and province capitals. The HYDE 3.2.1 keyframes are baked into `data/population/` (300 BC to 2017 AD, the 300 BC one derived; see MAP_DATA.md), so the engine runs on the real grid: 284,626 land nodes, loaded in ~0.5 s. The whole game is C# (ported from

GDScript with identical results): a yearly population tick takes ~30 ms and generating the world ~3 s. The old known issue (the non-player world lagging HYDE's total as history rose) is gone: regional totals now follow HYDE exactly.

Each phase should be independently verified against the real headless Godot engine (the Godot 4.7.2 .NET build; the session hook installs it) before moving to the next. The tests are under `src/Tests/`, one scene-tree script per suite, run with `godot --headless --path . --script res://src/Tests/<Name>.cs`; each exits with 1 if any check fails.